

# Graph Clustering approaches for Speaker Diarization of Conversational Speech

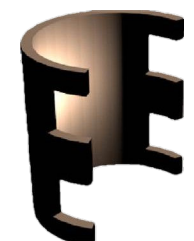
**PhD Thesis Defense**  
**6<sup>th</sup> February 2024**

**Prachi Singh**

**PhD Student,**

**Learning and Extraction of Acoustic Patterns (LEAP) Lab,  
Electrical Engineering, Indian Institute of Science, Bangalore.**

**Advisor: Dr. Sriram Ganapathy**



# Outline

---

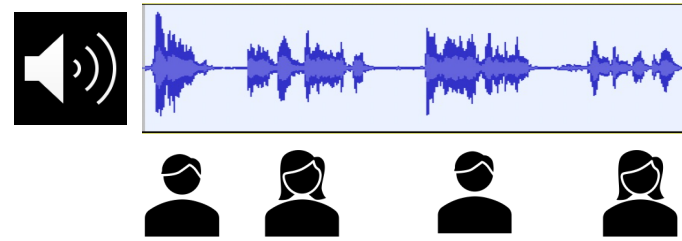
- Introduction
  - Motivation
  - Methodology
  - Contributions
- Background study
  - Related work
  - Performance metrics
  - Datasets
- Proposed Graph Clustering approaches
- Conclusion and Future Directions

# Introduction

---

# Motivation

Transcribing audio into text using speaker information generates much meaningful text



Hello

Hello. How are you Nitin?

I am doing great. How are you Meenu?

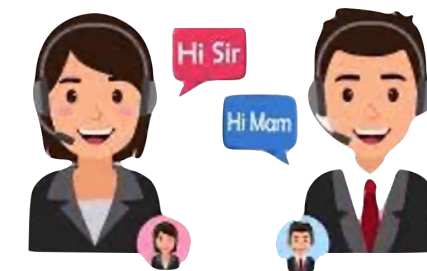
I am doing also great.

The task of finding “**who spoke when**” is called **Speaker Diarization**.

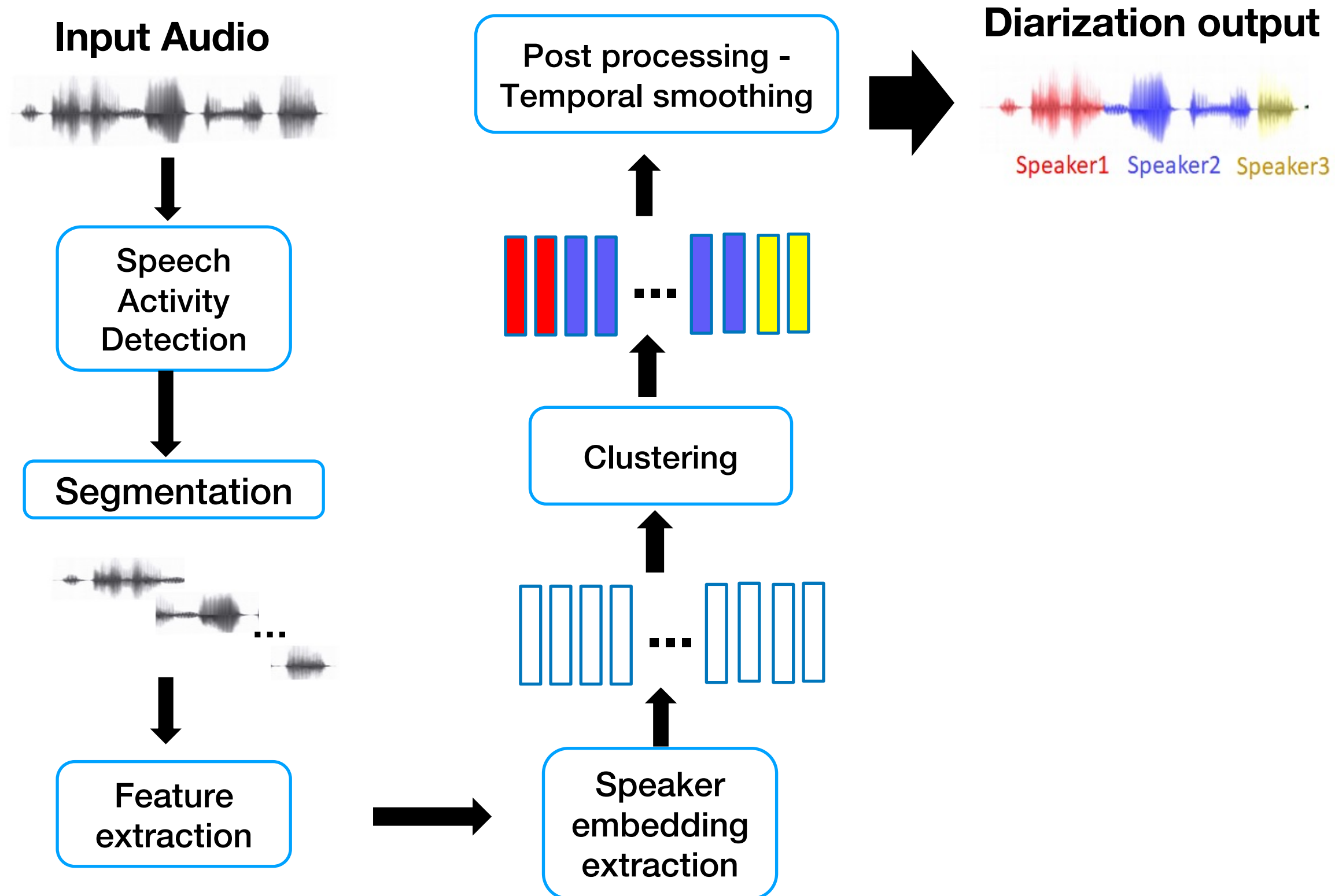
Transcribing meeting



Call center  
interactions  
Analysis

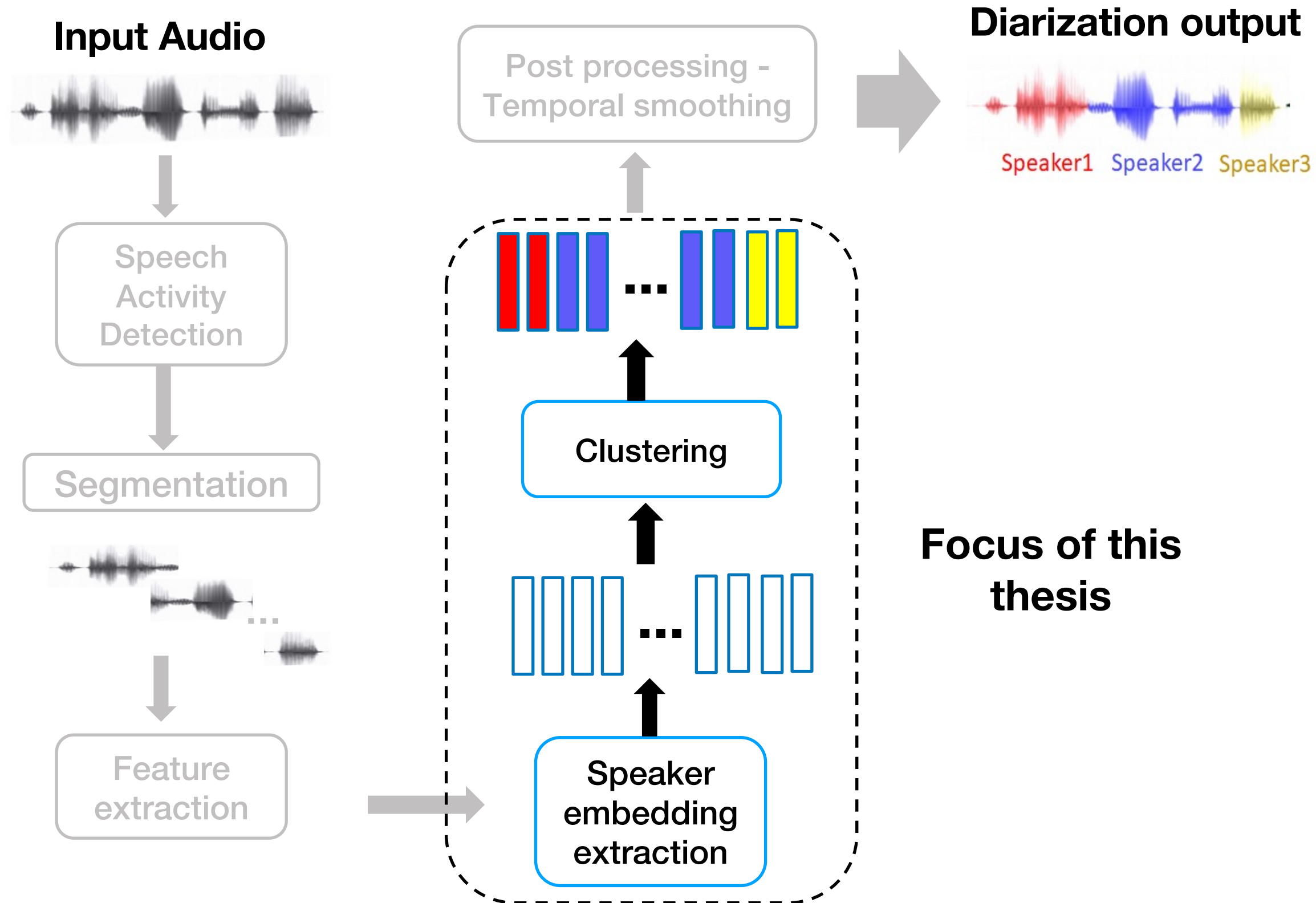


# Methodology



Sell et al., Diarization is hard: some experiences and lessons learned for the JHU team in the inaugural DIHARD challenge, 2018.

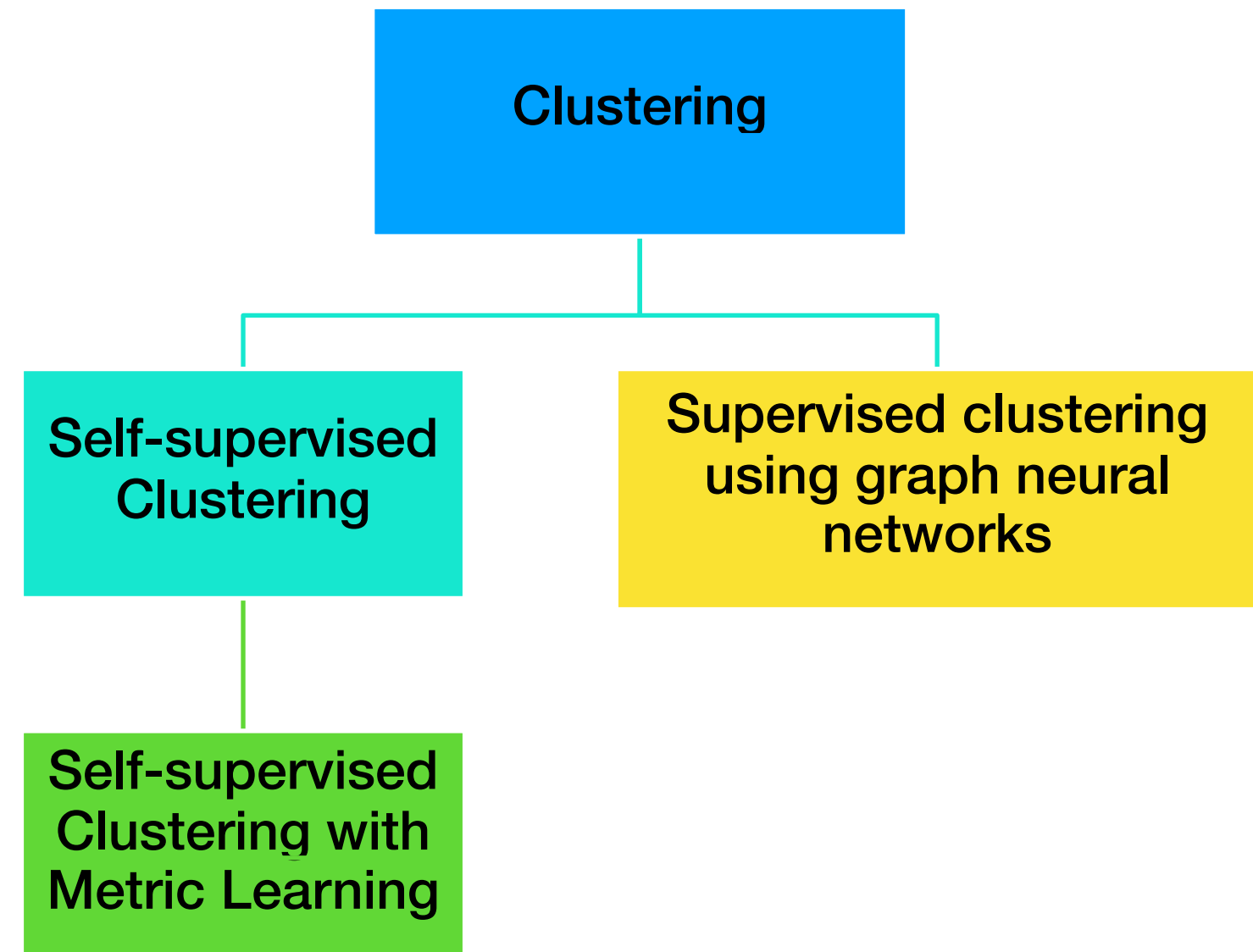
# Contributions outline



# Contributions outline

Application of **graph models** to **temporal segmentation of speech** is the first of its kind.

- Novel hierarchical graph clustering
- Self-supervised metric learning to generate similarity for clustering
- Supervised hierarchical graph clustering



# Background study

---

# Related work

---

## Unsupervised Clustering approaches

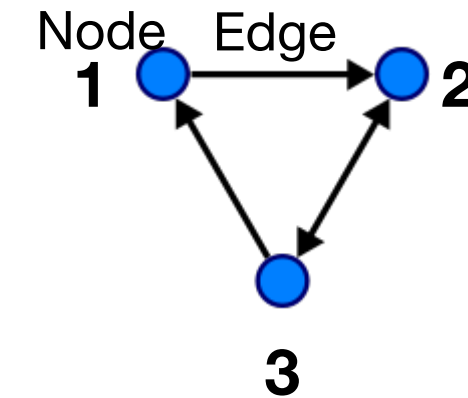
Forming groups based on hidden patterns in the unlabeled data

- Hierarchical clustering - Agglomerative Hierarchical Clustering (AHC)
- Graph Clustering

# Clustering approaches

## Graph

- A graph  $G$  can be well described by the set of **vertices**  $V$  and **edges**  $E$  it contains.  $G=(V,E)$
- The **vertices** are often called **nodes**.
- **Adjacency matrix (A)** captures connections between nodes,
- $A_{ij} = 1$ , if Node  $i$  is connected to  $j$  by an edge
- $A_{ij} = 0$ , if Node  $i$  and  $j$  are not connected
- A with real weights to the edges is called as weighted adjacency matrix.

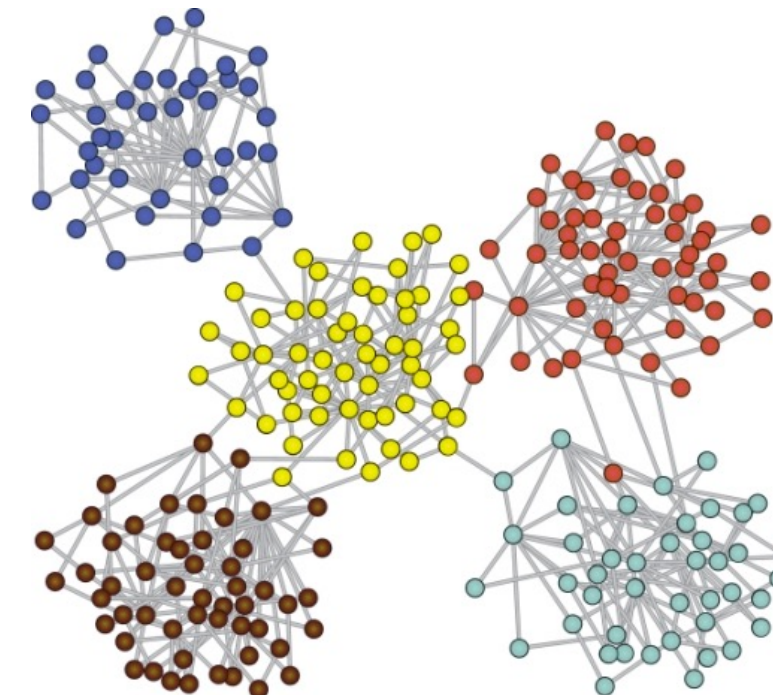


$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

## Graph clustering

Clustering the nodes such that **many edges** are present **within each cluster** and **fewer edges between the clusters**.

**Example: Spectral Clustering (SC)**



# Recent Graph Approaches

- **Graph attentional/convolution encoder (GAE)<sup>1</sup>** based approach for metric learning followed by spectral clustering.
- **Graph Attention- Based Deep Embedded Clustering (GADEC)<sup>2</sup>** : Graph attention-based clustering using multi-objective training.

<sup>3</sup>Wang et al., ICASSP, 2020

<sup>2</sup>Wei et al., Speech Communications, 2023

# Related work

- Speaker embeddings/representations
  - i-vector<sup>1</sup> – statistical model
  - d-vector<sup>2</sup> – Deep Neural Network
  - **x-vector**<sup>3</sup> – Time delay Neural Network (widely used)
- Similarity measure
  - Cosine<sup>4</sup>
  - **PLDA**<sup>5</sup> (widely used)

<sup>1</sup>Dehak et al., 2011, <sup>2</sup>Variani et al., 2014, <sup>3</sup>Snyder et.al.,2018

<sup>4</sup>Senoussaoui et al., 2014, <sup>5</sup>Sell and Garcia-Romero, 2014, <sup>5</sup>Sell et al., 2018

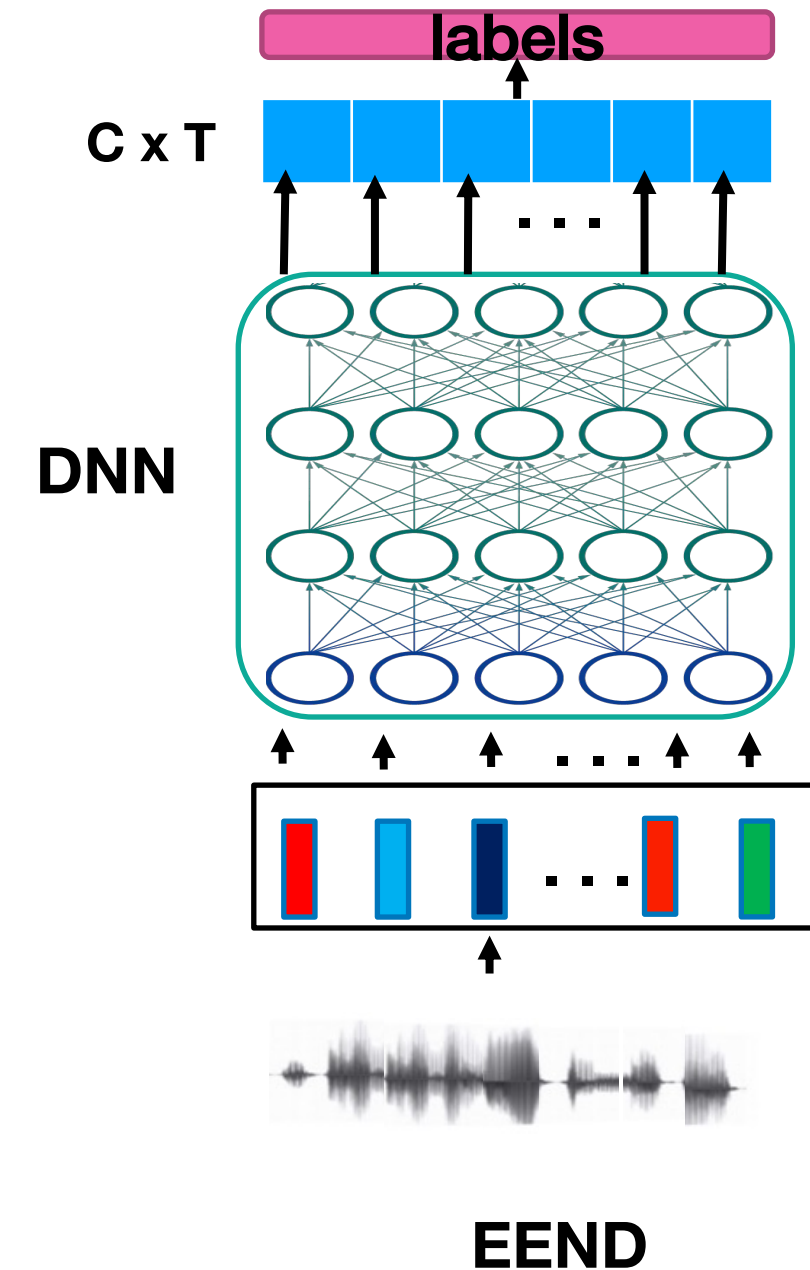
# Related work

## End to end neural diarization (EEND)<sup>1</sup>

- Transformer is used to perform speaker activity detection
- Takes input as F-dimensional audio features and generate C speaker labels

### Cons:

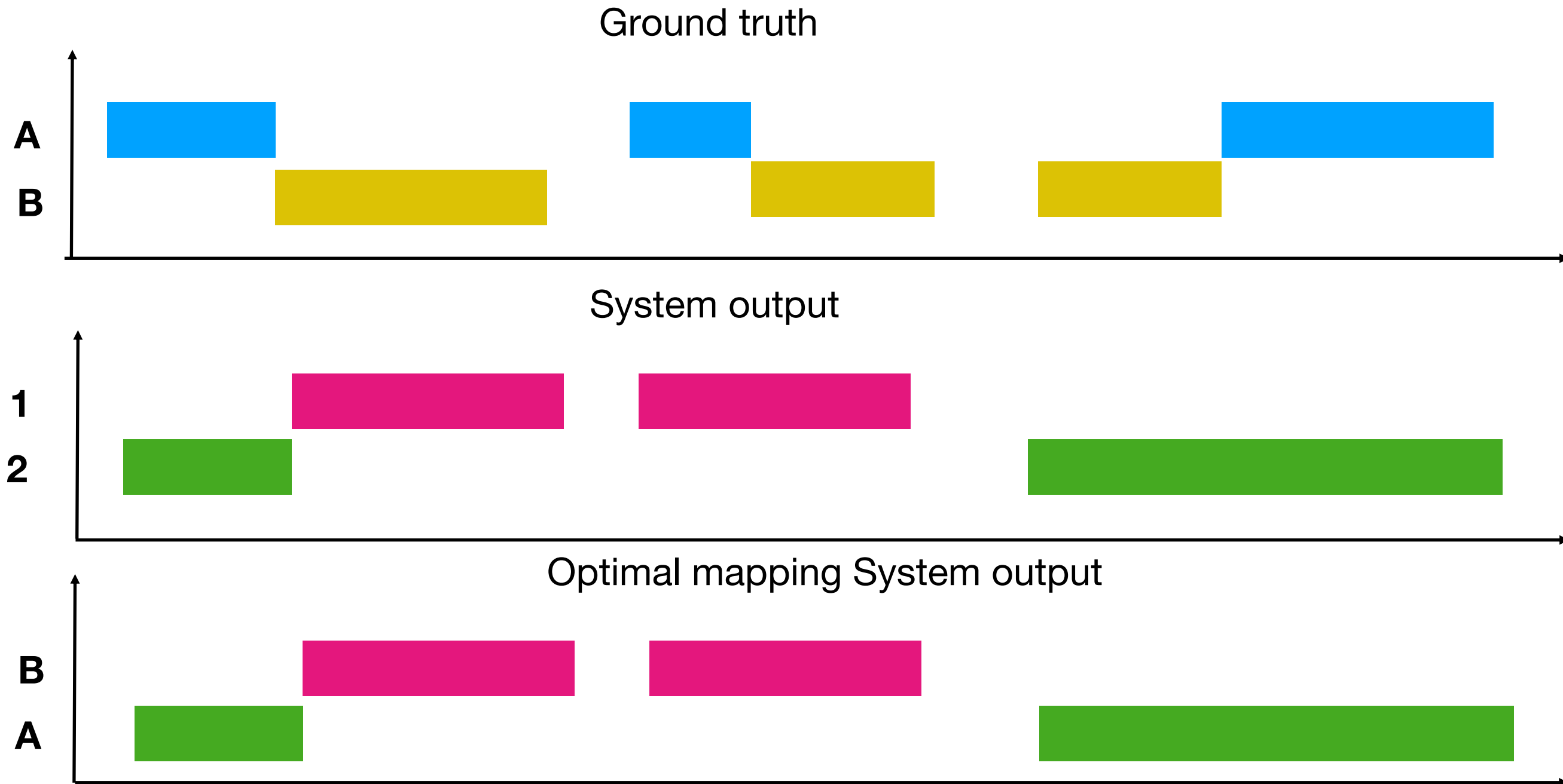
- Requires huge amount of labelled data for training.
- Difficult to generalize for higher number of speakers.
- Cannot handle long duration recording at a time.



<sup>1</sup>Horiguchi et. al., "End-to-End Speaker Diarization for an Unknown Number of Speakers with Encoder-Decoder Based Attractors

# Performance metric

Optimal mapping:  $\text{argmax}(A \cap 1, A \cap 2), \text{argmax}(B \cap 1, B \cap 2)$



# Performance metric

- **Diarization error rate (DER)** is the standard metric for evaluating and comparing speaker diarization systems.

- It is defined as follows:

$$DER = \frac{\text{false alarm} + \text{miss detection} + \text{speaker confusion}}{\text{total speakers duration}}$$

- *false alarm* - duration of non-speech predicted as speech
- *miss detection* - duration of speech of a speaker predicted as non-speech
- **speaker confusion** – duration of a reference speaker predicted as another speaker in system output after optimal mapping
- *total speakers duration* – total duration of all the speakers present

# Test Datasets

Narrowband  
(sampling rate: 8kHz)

## CALLHOME (CH) [1]

- Telephone data
- #Recordings - 500
- Duration: 2-5 mins
- 2-7 speakers

Wideband (sampling rate: 16kHz)

## AMI [2]

- Meeting dataset
- #Recordings - Dev set: 18, Eval set: 16
- Duration: 20-60mins
- 3-5 speakers

## DIHARD III [3]

- Speech diarization challenge data
- 9-11 domains
- #Recordings – Dev set:254, Evalset:259
- Duration: 0.5-10 mins
- 1-10 speakers

## Voxconverse [4]

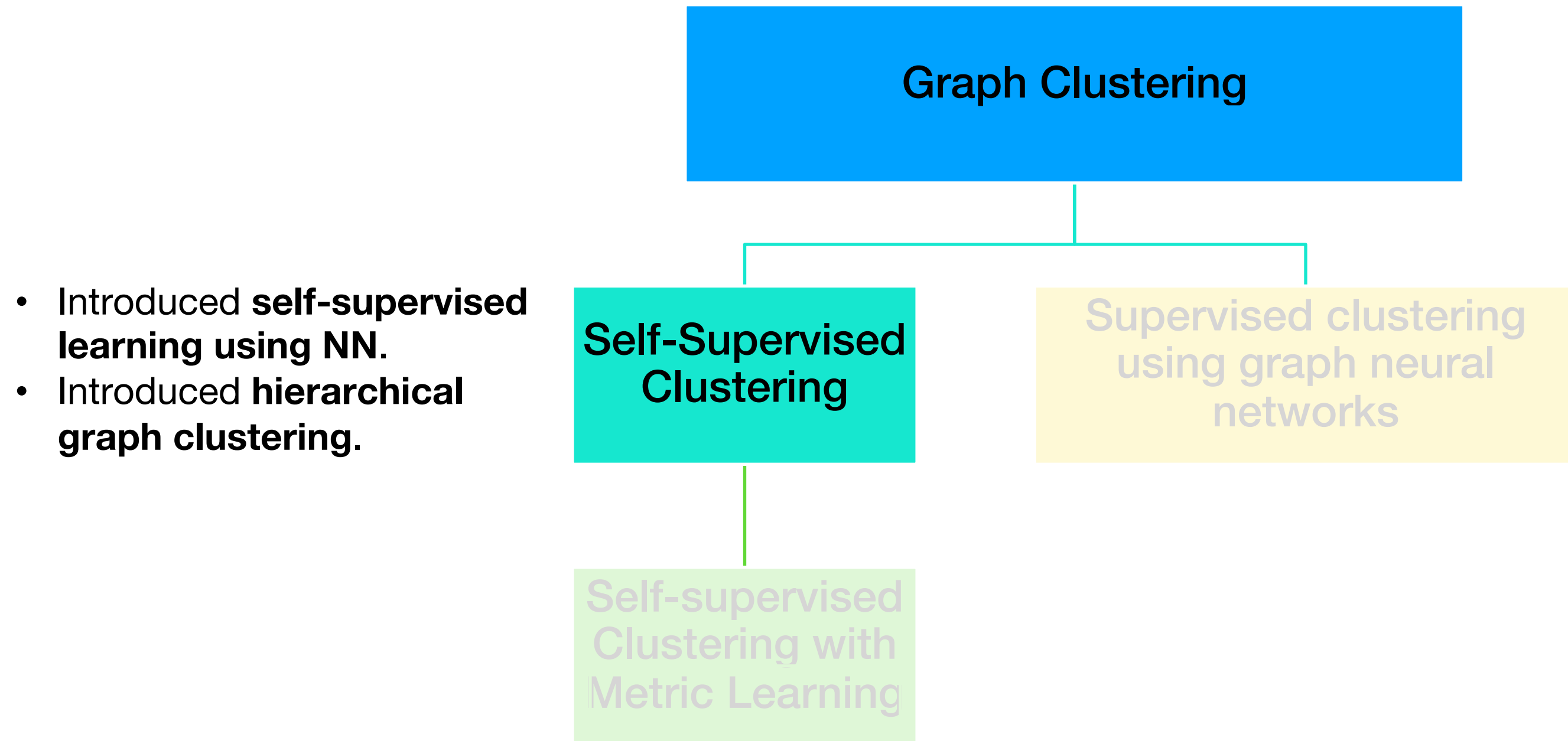
- ]extracted from YouTube videos.
- #Recordings - dev set: 216 and eval set: 232
- Duration: 22s - 20mins.
- 1-21 speakers.

[1] Mark et al., 2000 NIST Speaker Recognition Evaluation

[2] McCowan et al., The AMI meeting corpus, 2005

[3] Ryant et al., The Third DIHARD Diarization Challenge, 2020

[4] Chung et al., Spot the Conversation: Speaker Diarisation in the Wild, 2020



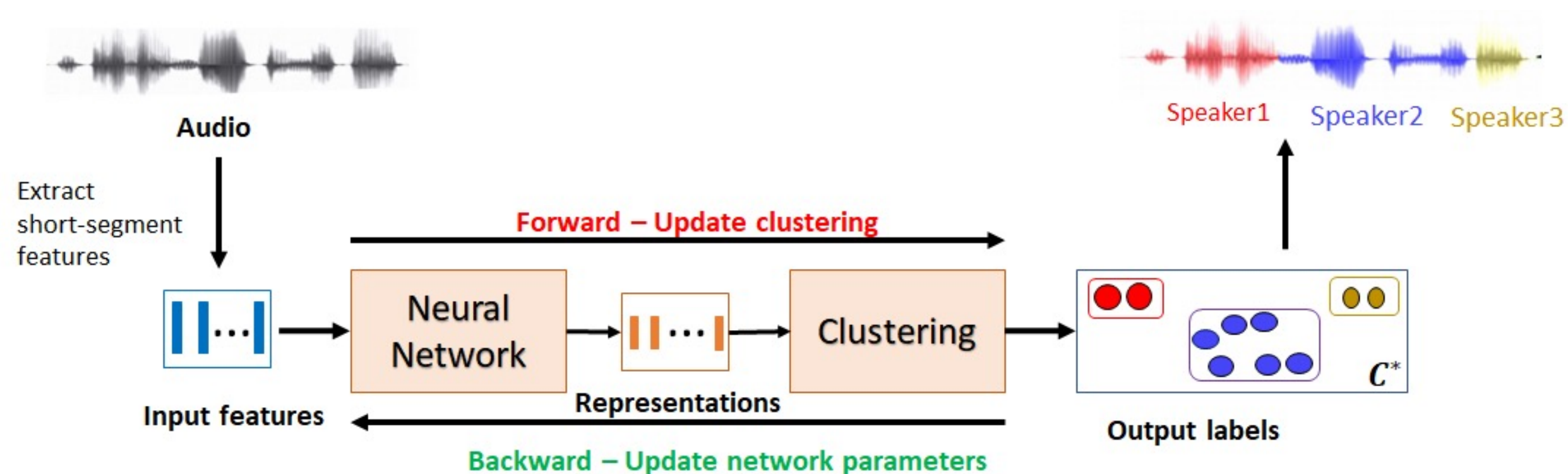
## Proposed Approach 1

# Motivation

- Each stage (embedding extraction and clustering) is **optimized independently**.
- The test set will contain unseen domains and speakers.
- **Can clustering** provide self-supervisory targets for representation learning<sup>1</sup>?
- Can we improve clustering using the **succinct speaker representations** ?

# Self-supervised clustering

**Self-Supervised Clustering** alternates between **merging the clusters** for a fixed embedding representation and **learning the representations** using the given cluster labels, till we reach the **required number of clusters/speakers**.



Prachi Singh, Sriram Ganapathy, 'Deep self-supervised hierarchical clustering for speaker diarization', INTERSPEECH 2020.

# SSC Algorithm

## Variables:

$\mathbf{X} = \{x_1, \dots, x_{N_r}\} \in R^D$ : X-vectors sequence of recording  $r$

$\mathbf{Y} = \{y_1, \dots, y_{N_r}\} \in R^d$ : lower dimensional representations

$\mathbf{z} = \{z_1, \dots, z_{N_r}\} \in R$ : segment labels

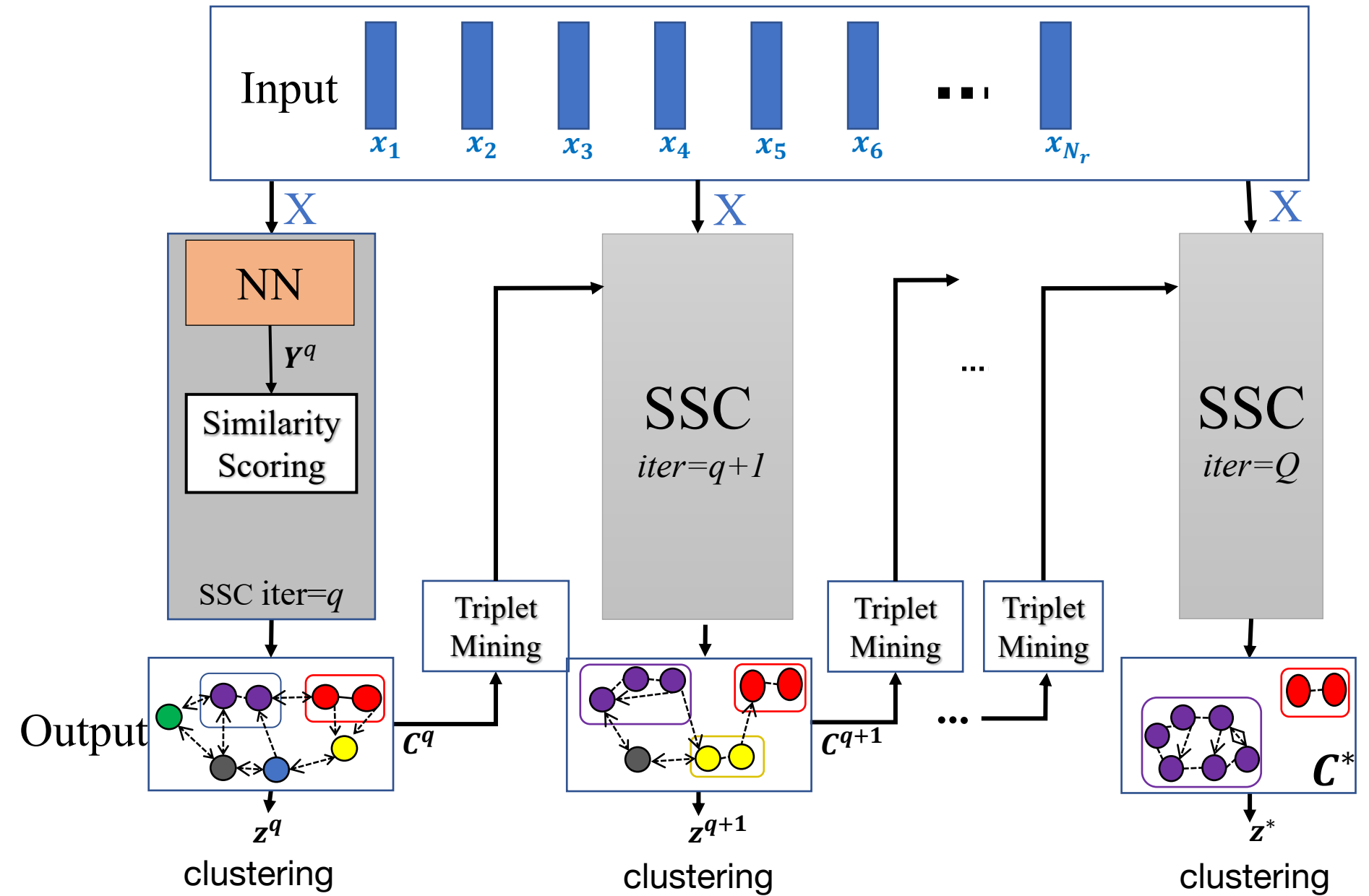
$\theta$ : DNN parameters

$(\mathbf{Y}^q, \mathbf{z}^q, \theta^q)$ : refer to variables at iteration  $q$

$N^q$ : Number of clusters at iteration  $q$

$N^*$ : target number of clusters

For NN training at iteration  $q$ , use clustering results from  $q-1$  to sample positive and negative pairs of triplets.



P. Singh and S. Ganapathy, "Self-supervised Representation Learning With Path Integral Clustering For Speaker Diarization", IEEE TASLP (2021).

# NN training–Triplet loss

- For each cluster  $\mathcal{C}_i^q$ , pick two elements as anchor and positive  $\{y_i, y_j\}$ .
- For negative pair, element ( $y_l$ ) is selected randomly from any other cluster.
- Triplet loss:

$$\theta^q = \operatorname{argmax}_{\theta} \sum_{i,j,l} [s(i,j) - \alpha(s(i,l) + s(j,l))]$$

$s(i,j)$  – similarity score ;  $0 < \alpha \leq 1$  : weighting factor

# Agglomerative clustering

## AHC

Merging Criterion:

In an AHC algorithm, the merging criterion for merging two clusters  $C_a^q$  and  $C_b^q$  where  $q$  is the iteration index is given as

$$\{C_a^q, C_b^q\} = \arg \max_{C_i, C_j \in C, i \neq j} A(C_i, C_j)$$

(where,  $A$  denote the affinity measure between two clusters.)

# Agglomerative clustering

## Path integral clustering (PIC)

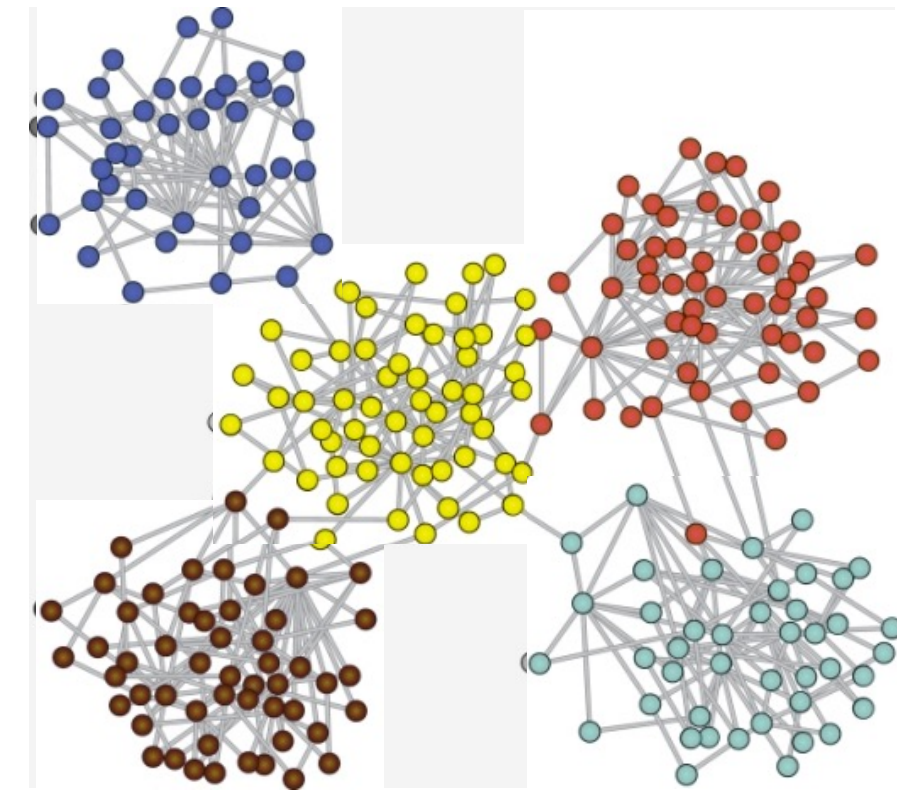
Graph-structural based agglomerative clustering algorithm where graph encodes the structure of the embedding space.

1. **Measures the affinity of clusters based on the neighborhood graph hence is more robust to noisy distances.**
2. **Uses robust graph structural merging strategy for noisy links.**
3. **It does not assume anything on the underlying data distributions and only need the pairwise similarities of samples.**

# Path Integral Clustering (PIC)

- Given a set of vectors  $X = \{x_1, x_2, \dots, x_n\}$ , it involves creation of directed graph  $G = (V, E)$
- Weighted Graph Adjacency matrix (W) given as,  
$$w_{ij} = S(i, j) \text{ if } x_j \in N_i^K$$
$$= 0 \text{ otherwise}$$

where,  $S(i, j)$  is the pairwise similarity between  $x_i$  and  $x_j$ ,  $N_i^K$  is the set of K nearest neighbour of  $x_i$



# Baselines<sup>12</sup>

| Step                                                | Parameter             | CH                | AMI               |
|-----------------------------------------------------|-----------------------|-------------------|-------------------|
| -                                                   | Sampling rate         | 8kHz              | 16kHz             |
| Segmentation                                        | Window size           | 1.5s, 0.75s shift | 1.5s, 0.75s shift |
| Embedding<br>extraction<br>(x-vector)<br>extraction | Architecture          | 7-layers TDNN     | 7-layers TDNN     |
|                                                     | Train set             | SWBD, SRE         | Voxceleb 1,2      |
|                                                     | Train #speakers       | 4,285             | 7,323             |
|                                                     | Input features        | 23D MFCCs         | 30D MFCCs         |
|                                                     | x-vector<br>dimension | 128               | 512               |
| Similarity score                                    | type                  | PLDA              | PLDA              |
| Clustering                                          | type                  | AHC               | AHC               |

# Implementation details

| config              | CH     | AMI        |
|---------------------|--------|------------|
| x-vectors/recording | 50-700 | 1000-4000  |
| 2-layer DNN         | 128x10 | 512X30     |
| Learning rate       | 0.001  | 0.001      |
| Annealing           | No     | Yes        |
| Batch               | Full   | Mini-batch |
| epochs              | 5-10   | 5-10       |

# Initialization

- **Weight initialization** and **training** are **file specific**.
- Uses **processing steps** from **baseline system**.
- **First layer** is initialized using **global PCA** computed using held out set followed by **length norm**.
- **Second layer** is initialized using **file-level PCA**.
- Affinity measure : **Cosine** similarity.

# CH Results

- Performance metric: Diarization Error Rate (DER) (%)
- Considering only non-overlapping speech regions with tolerance collar (0.25s).

| System               | Known N*   | Unknown N* |
|----------------------|------------|------------|
| x-vec + cosine + AHC | 8.9        | 10.0       |
| x-vec + cosine + SC  | 9.4        | 11.9       |
| x-vec + PLDA + AHC   | 7.0        | 8.0        |
| x-vec + cosine + PIC | 7.7        | 9.3        |
| SSC-AHC              | 6.4        | 8.3        |
| SSC-PIC              | 6.4        | 7.5        |
| + Temp. cont.        | <b>6.3</b> | <b>7.0</b> |

# AMI Results

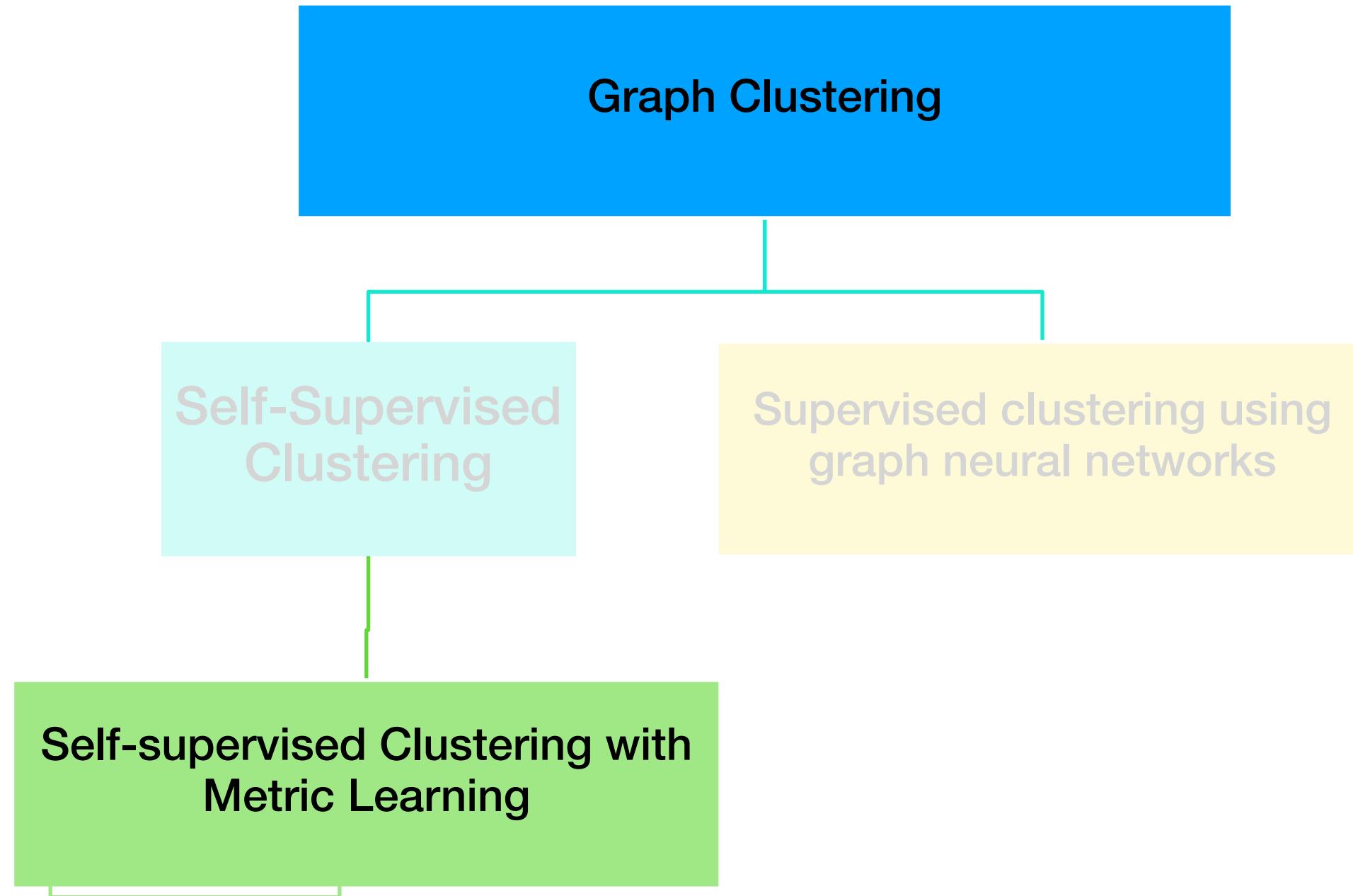
| System                           | Known N*   |            | Unknown N* |            |
|----------------------------------|------------|------------|------------|------------|
|                                  | Dev.       | Eval.      | Dev.       | Eval.      |
| x-vec + cosine + AHC             | 34.6       | 30.2       | 18.2       | 15.5       |
| x-vec + cosine +SC               | 30.2       | 25.5       | 40.0       | 31.1       |
| x-vec + PLDA + AHC<br>(Baseline) | 15.7       | 16.0       | 13.7       | 16.3       |
| SSC-PLDA-AHC                     | 9.4        | 11.1       | 10.7       | 11.6       |
| x-vec + PLDA + PIC               | 9.4        | 9.3        | 9.8        | 10.4       |
| x-vec + cosine + PIC             | 8.9        | 7.3        | 9.0        | 7.3        |
| SSC-PIC                          | 7.3        | 7.2        | 8.1        | 7.6        |
| + Temp. cont.                    | <b>6.2</b> | <b>6.4</b> | <b>6.4</b> | <b>6.7</b> |

Prachi Singh and Sriram Ganapathy, “Self-supervised Representation Learning with Path Integral Clustering for Speaker Diarization”, IEEE Transactions on Audio Speech and Language Processing, 2021.

# Summary

---

- Proposed **self-supervised clustering algorithm using DNN** which iteratively updates representation learning and clustering.
- Introduced **path integral clustering – hierarchical graph clustering for first time for diarization.**
- Helped to **increase separation between representations** of different speakers.
- Showed improvements on AMI and CALLHOME dataset.



## Proposed Approach 2

# Motivation

- **SSC uses cosine similarity** to perform clustering.
- **Prior work on clustering performs better with PLDA score than cosine.**
- PLDA<sup>1</sup> is a parametric model which is trained to learn speaker distributions.
- Can we train the SSC with learnable scoring/metric function?

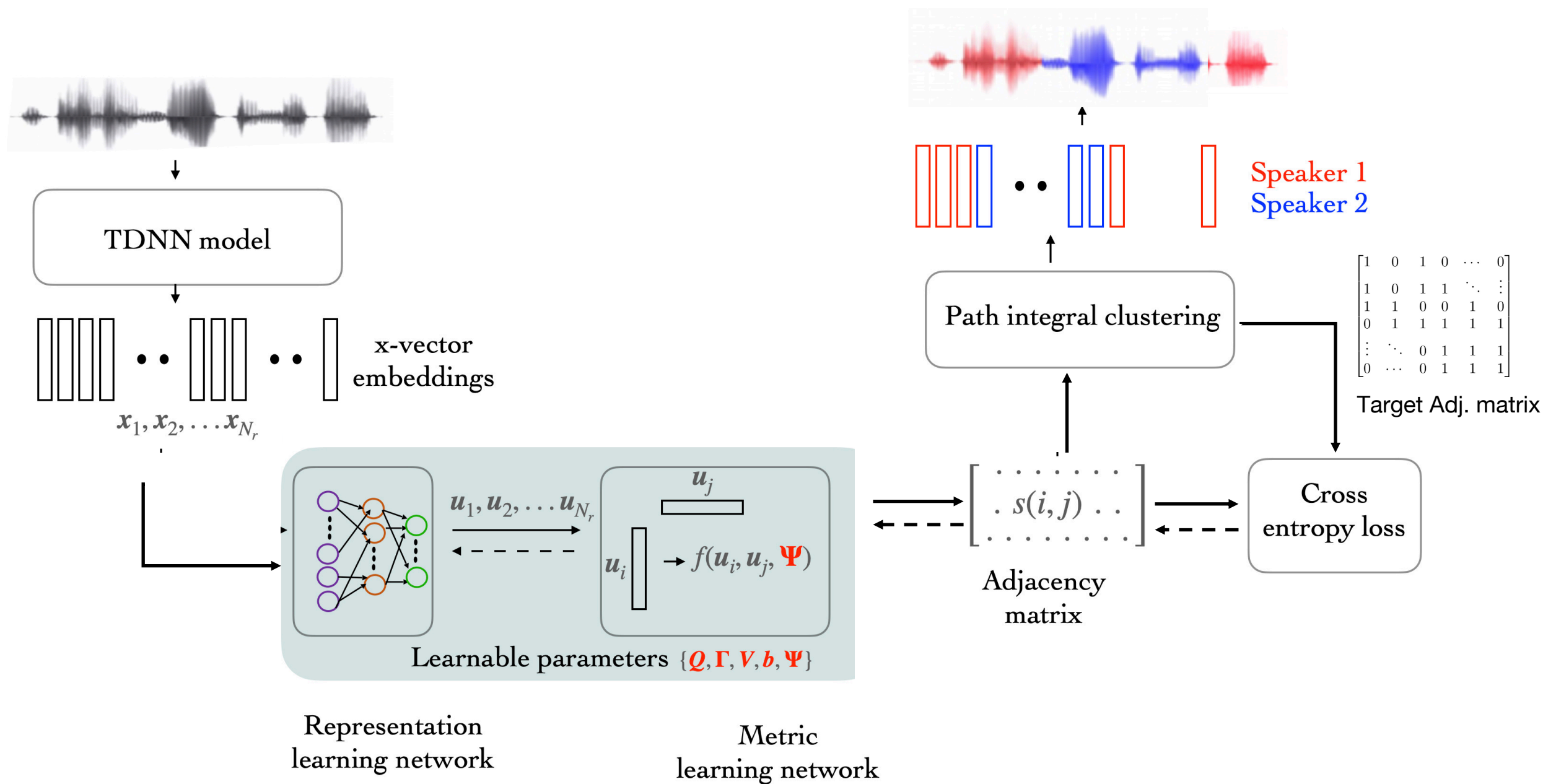
<sup>1</sup>Sergey Ioffe, Probabilistic linear discriminant analysis, 2006

# SelfSup-PLDA-PIC

- **Self-supervised metric learning with graph-based clustering algorithm (SelfSup-PLDA-PIC) jointly performs representation learning and metric learning** using the initial clustering results.
- **Propose a neural version of PLDA** to incorporate **deep learning** of the **PLDA** model parameters.

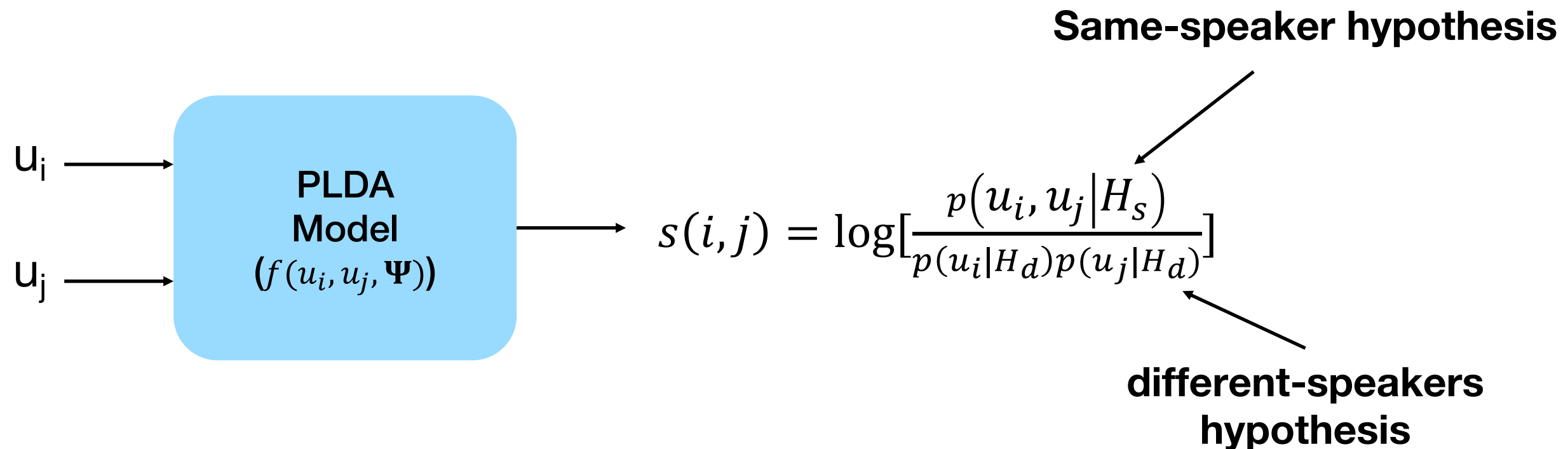
P. Singh and S. Ganapathy, “Self-Supervised Metric Learning with Graph Clustering for Speaker Diarization”, IEEE ASRU 2021.

# Block diagram: SelfSup-PLDA-PIC



# Metric Learning using PLDA model

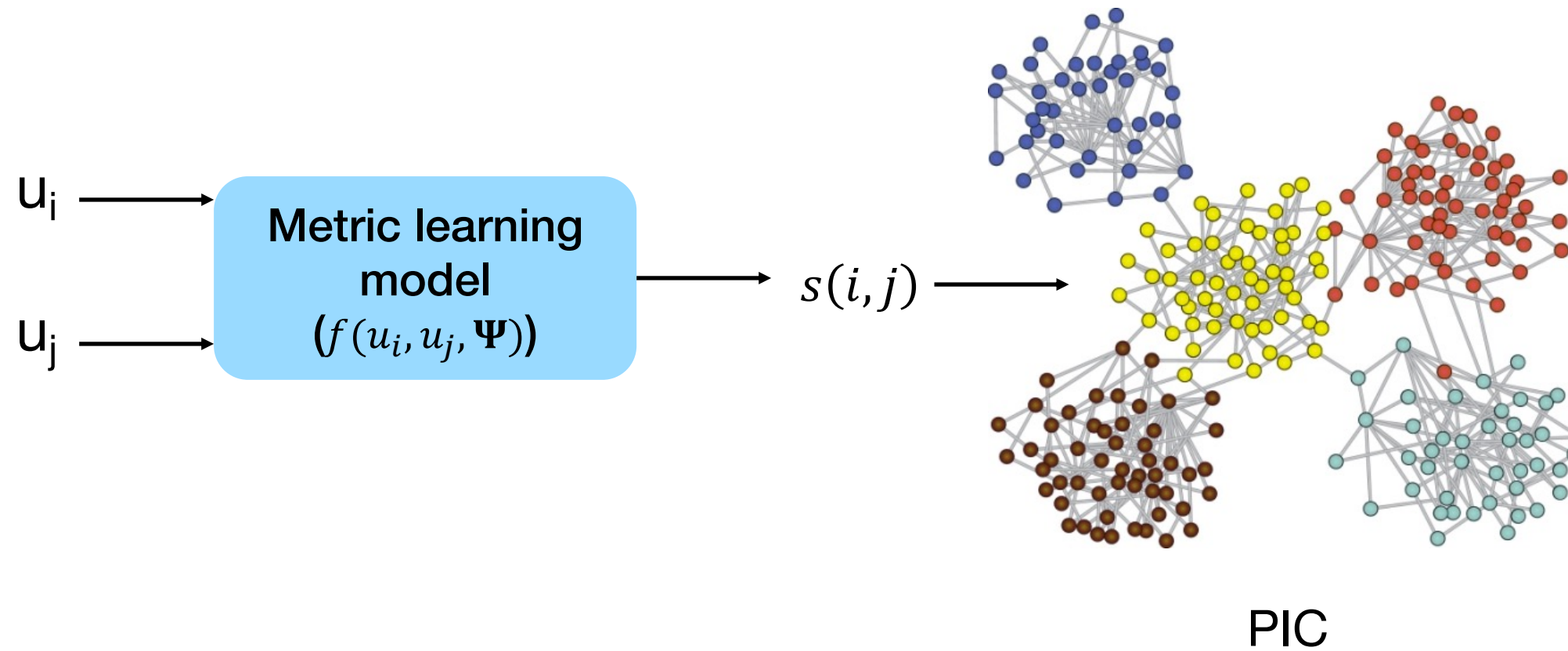
- Probabilistic Linear Discriminant Analysis (PLDA)<sup>1</sup> is a supervised generative model trained to learn distributions of different speakers.
- It can be used to find pairwise similarity score between embeddings from unseen speakers as follows



<sup>1</sup>Sergey Ioffe, Probabilistic linear discriminant analysis, 2006

# Metric Learning using PLDA model

- Replacing PLDA model with a learnable parametric model with parameter  $\Psi$



# AMI Results

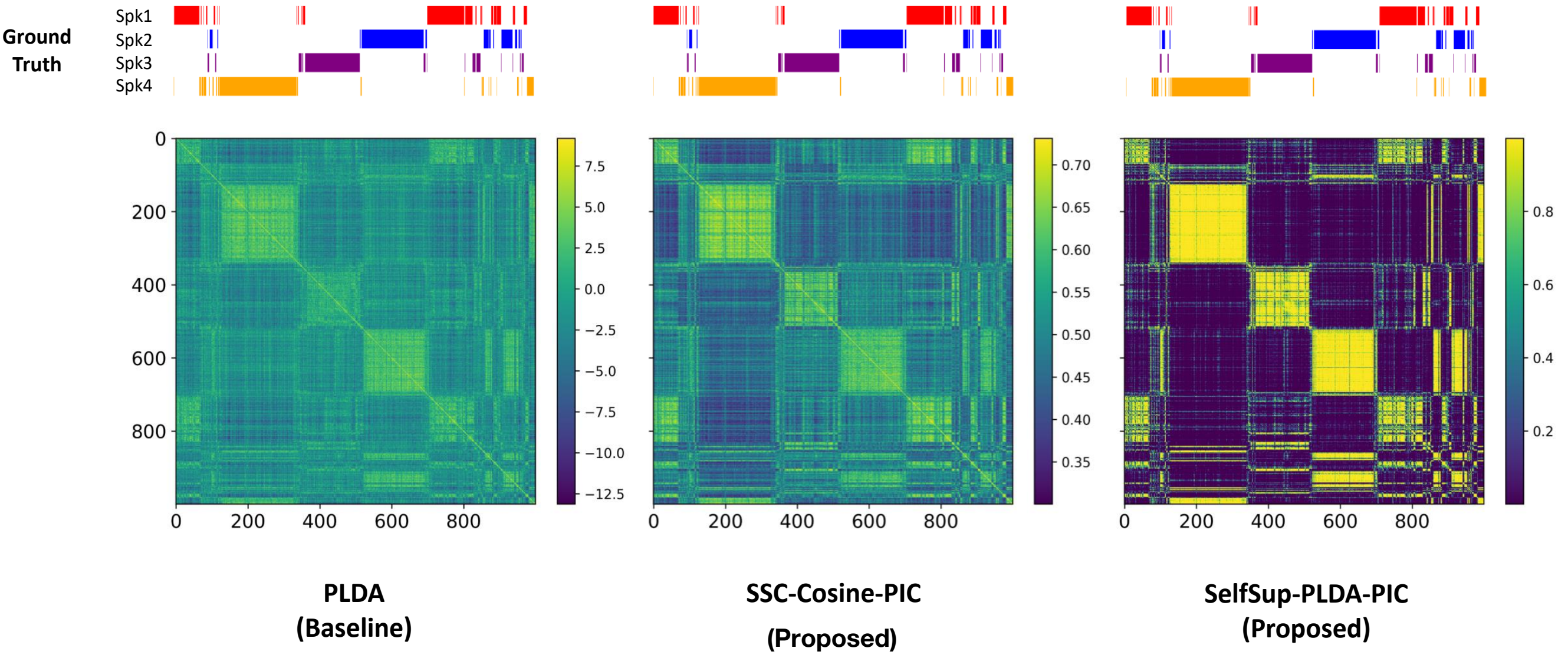
**AMI DER (%) Results** – Ignoring overlaps and with collar 0.25s

| System                                    | Known N*    |             | Unknown N*  |             |
|-------------------------------------------|-------------|-------------|-------------|-------------|
|                                           | Dev.        | Eval.       | Dev.        | Eval.       |
| <b>x-vec + PLDA + AHC</b>                 | <b>15.9</b> | <b>12.2</b> | <b>13.1</b> | <b>12.3</b> |
| x-vec + PLDA + PIC                        | 5.1         | 10.2        | 5.8         | 11.4        |
| <b>SSC-Cosine-PIC</b>                     | <b>5.3</b>  | <b>6.2</b>  | <b>6.5</b>  | <b>8.4</b>  |
| SelfSup-PLDA-AHC                          | 7.9         | 7.3         | 7.7         | 9.4         |
| <b>SelfSup-PLDA-PIC<sup>1</sup></b>       | 4.2         | 6.2         | 4.4         | 6.9         |
| <b>_ + Temporal continuity</b>            | 4.2         | 4.2         | <b>4.4</b>  | <b>4.9</b>  |
| <b>SelfSup-PLDA-PIC + VBx<sup>2</sup></b> | -           | -           | <b>2.9</b>  | <b>4.2</b>  |

<sup>1</sup>P. Singh and S. Ganapathy, “Self-Supervised Metric Learning with Graph Clustering for Speaker Diarization”, IEEE ASRU 2021.

<sup>2</sup>Diez et al., Bayesian HMM based x-vector clustering for speaker diarization, 2019

# AMI Visualization



Similarity score matrices comparison for 4-speaker recording from AMI development set

# DIHARD Results

**Average DER (%)** on the DIHARD dataset considering overlapping regions with no tolerance collar.

For recordings with  $\leq 7$  speakers and  $> 7$  speakers.

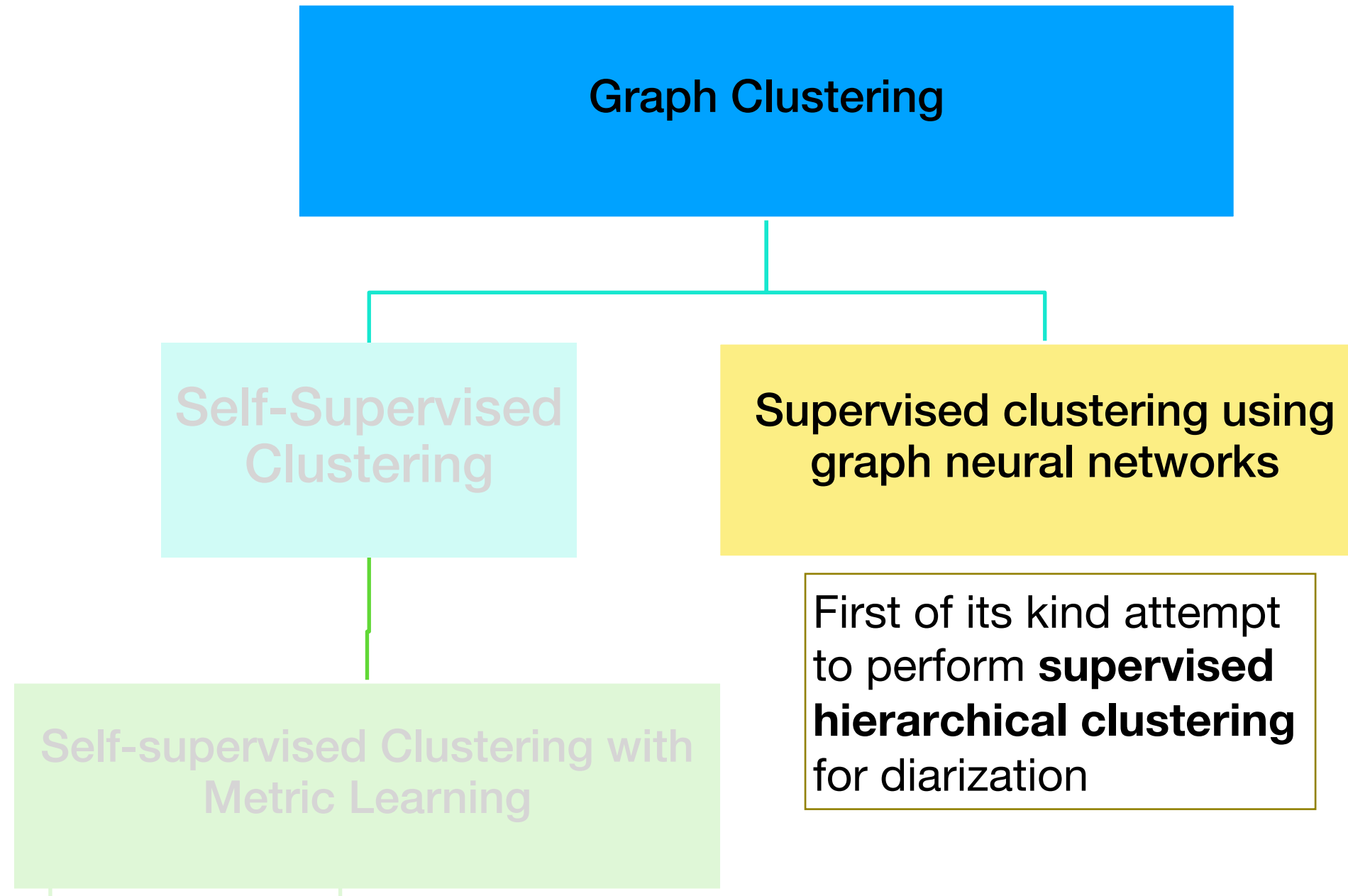
| System             | $\leq 7$ speakers |       | $> 7$ speakers |       |
|--------------------|-------------------|-------|----------------|-------|
|                    | Dev.              | Eval. | Dev.           | Eval. |
| X-vec + PLDA + AHC | 18.0              | 19.3  | 36.6           | 27.1  |
| X-vec + PLDA + PIC | 17.7              | 17.8  | 36.5           | 24.0  |
| SelfSup-PLDA-PIC   | 17.0              | 17.2  | 39.5           | 28.1  |

Performance degraded as number of speakers increases as initial clustering becomes unreliable.

# Summary

---

- Proposed **self-supervised metric learning** approach using **PLDA**.
- Adapted similarity scores for each test recording.



## Proposed Approach 3

# Motivation

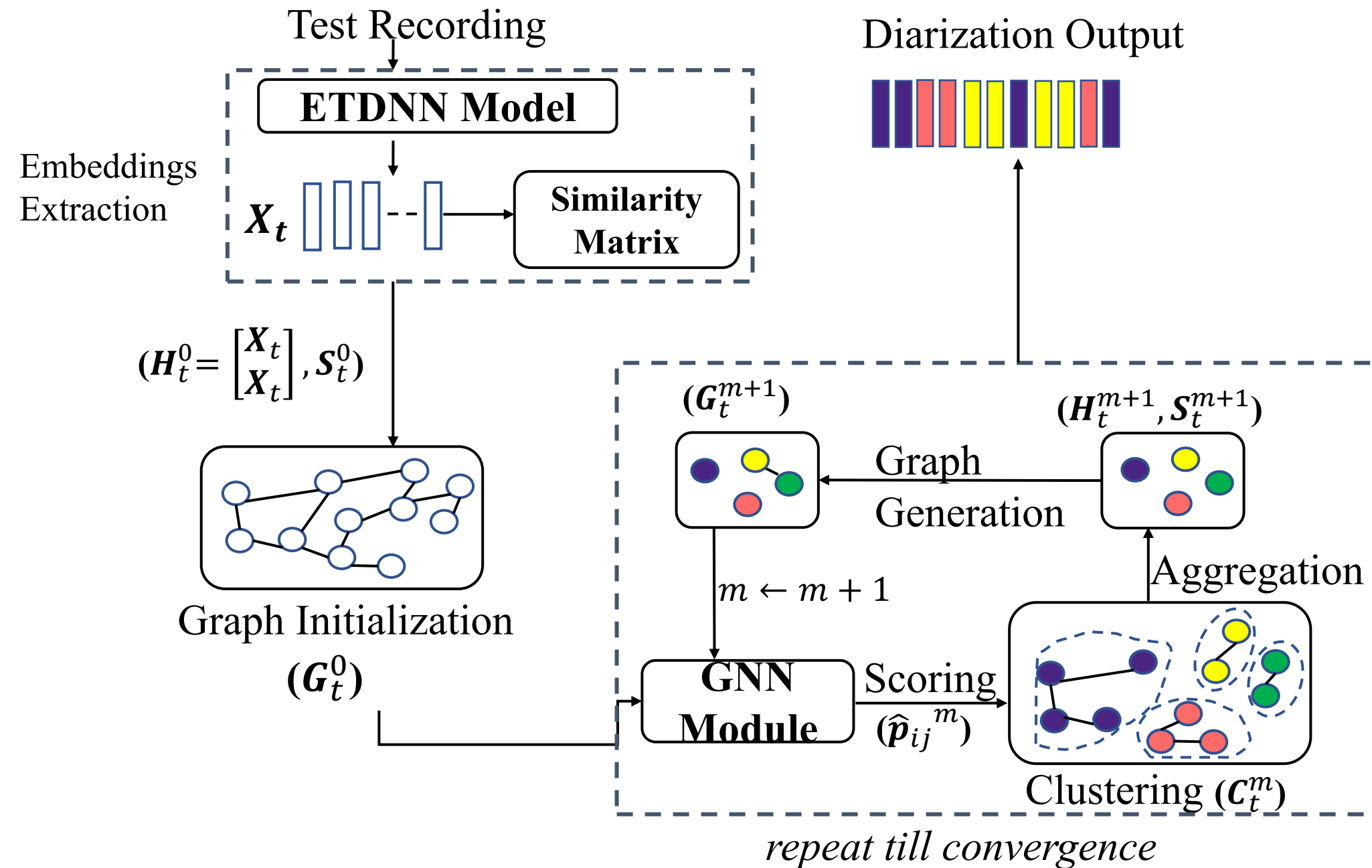
- Self-supervised clustering is less reliable when recording **contains higher number of speakers (>7)**.
- The end goal is **to minimize the clustering errors** to improve performance
- **Can we train a supervised model with the clustering objective?**

# Supervised Hierarchical Graph Clustering (SHARC)

- Performs **supervised clustering** using **Graph Neural Networks (GNN)**.
- Represents the speaker embeddings using graph.
- Clustering loss is used to update edges of the graph.
- Generates node labels based on clustering performed on updated edges at **each level of hierarchy**.
- **E-SHARC** : Joint learning of embedding extractor and GNN

Yifan Xing et. al, "Learning hierarchical graph neural networks for image clustering," in Proc. IEEE ICCV, 2021.

# Block diagram: E-SHARC Inference



# GNN scoring

- GNN scoring function  $\Psi$  - a learnable GNN module designed for supervised clustering.
- Output: edge prediction probability  $p_{ij}$  between node  $i$  and  $j$ .
- $N_i^k$  -  $k$ -nearest neighbors of node  $v_i$ ,

$$\hat{e}_{ij} = 2p_{ij} - 1 \in [-1, 1] \forall j \in N_i^k$$

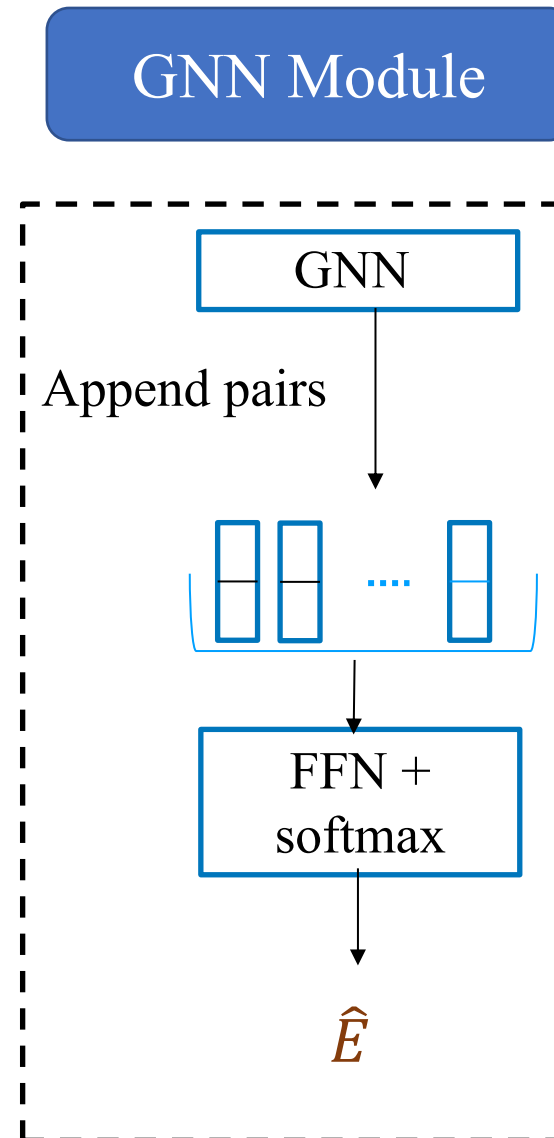
- Density of node  $i$  :

$$d_i = \frac{1}{k} \sum_{j \in N_i^k} e_{ij} \mathbf{S}_r(i, j)$$

- Ground truth:

$$\hat{d}_i = \frac{1}{k} \sum_{j \in N_i^k} \hat{e}_{ij} \mathbf{S}_r(i, j)$$

- Predicted:



# Clustering

- At each level of hierarchy  $m$ , it creates a candidate edge set  $\varepsilon(i)$

$$\varepsilon(i) = \{j | (v_i, v_j) \in E_m, \quad \hat{d}_i \leq \hat{d}_j \quad \text{and} \quad p_{ij} \geq p_\tau\}$$

- For any  $i$ , if  $\varepsilon(i)$  is not empty, we pick  $j = \operatorname{argmax}_{j \in \varepsilon(i)} \hat{e}_{ij}$
- A set of connected components  $C_t^m$ , forms clusters for the next level ( $m + 1$ ).



# Training loss

- Loss:  $L = L_{conn} + L_{den}$

- $L_{conn} = \frac{1}{|E|} \sum_{i,j \in E} p_{ij} \log \hat{p}_{ij} + (1 - p_{ij}) \log (1 - \hat{p}_{ij})$

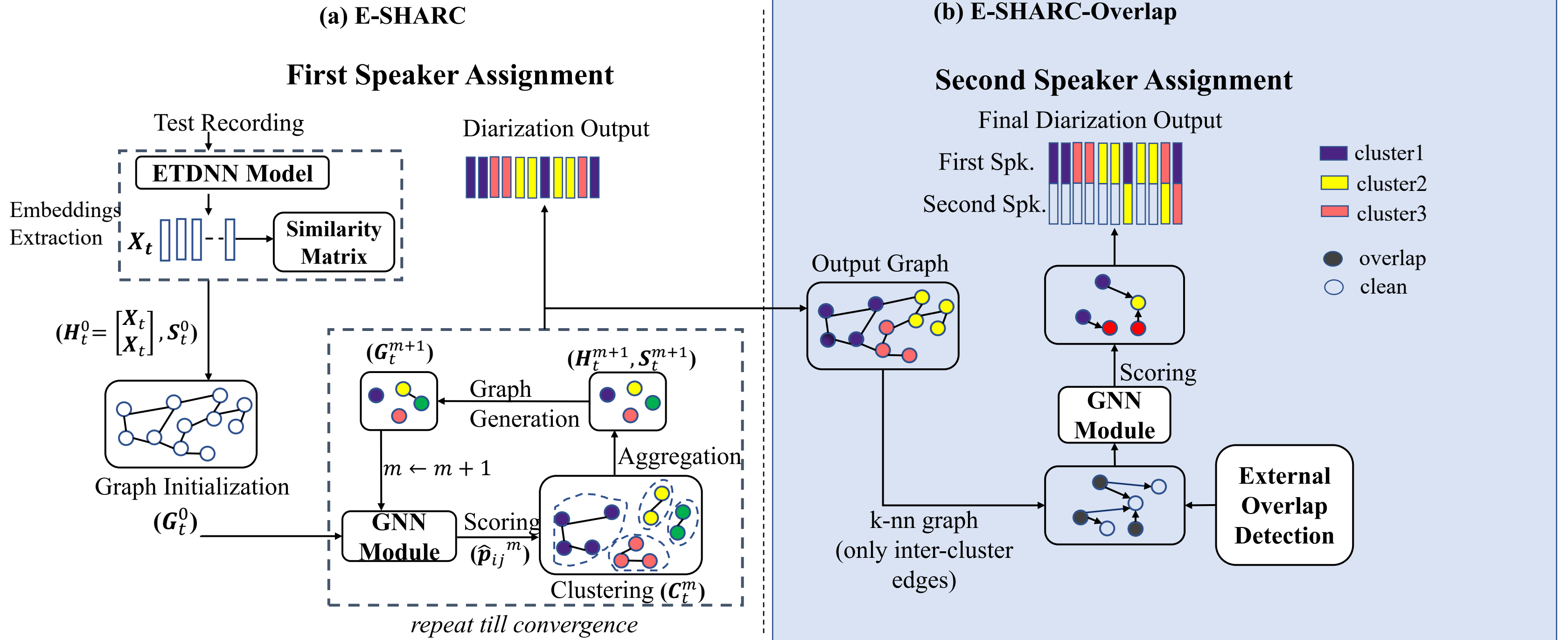
$p_{ij}$ - Ground truth edge labels,  $\hat{p}_{ij}$ - predicted edge labels

- $L_{den} = \frac{1}{|V|} \sum_{i=1}^{|V|} ||d_i - \hat{d}_i||_2^2 \quad \forall i \in \{1, \dots, |V|\}, \text{ where } |V| \text{ is the cardinality of } V$

- $d_i$  : ground truth node density,  $\hat{d}_i$ : predicted node density

- 
- What about overlapping speech ?

# Block diagram: E-SHARC-Overlap Inference



# Experiments

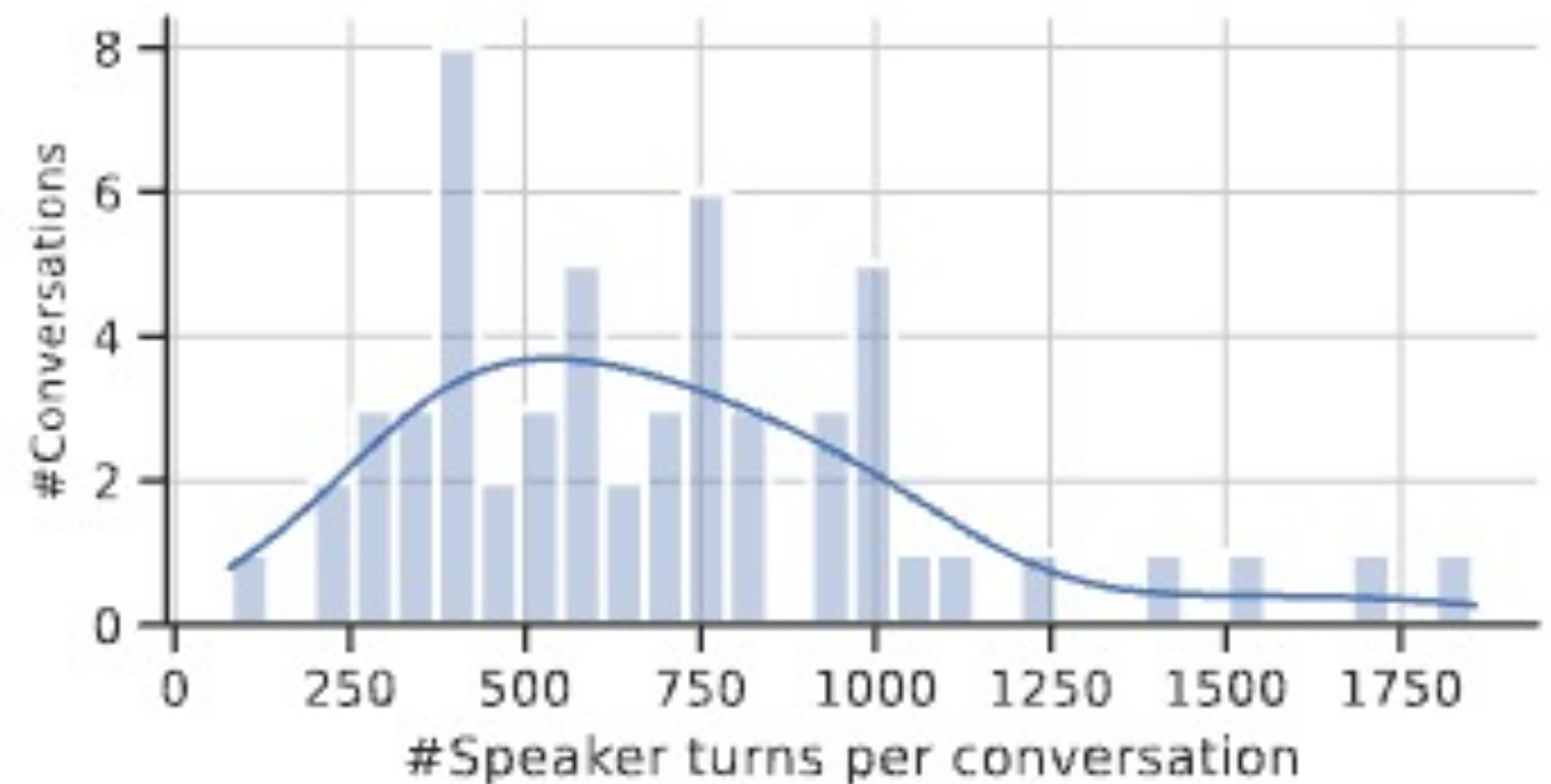
## Datasets

**AMI** : Meeting dataset

**Voxconverse**: Youtube videos

**DISPLACE 2023 dataset**<sup>1</sup>: Natural multilingual, multi-speaker speech recordings.

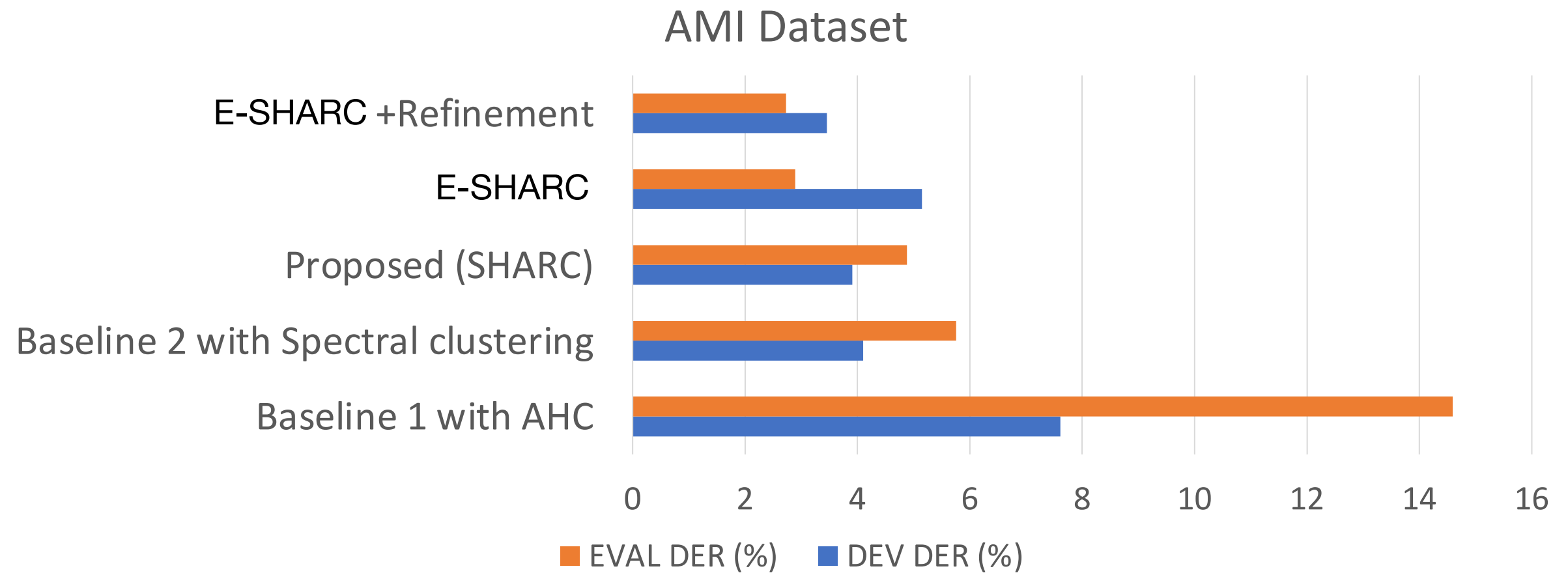
- #Recordings- dev set: 27 and eval set: 29.
- Duration: 30-60 mins
- #speakers varies from 3-5, and #languages varies from 1-3.



<sup>1</sup>Baghel et al., Interspeech 2023

# Results

- Performance : DER (%) (lower the better)

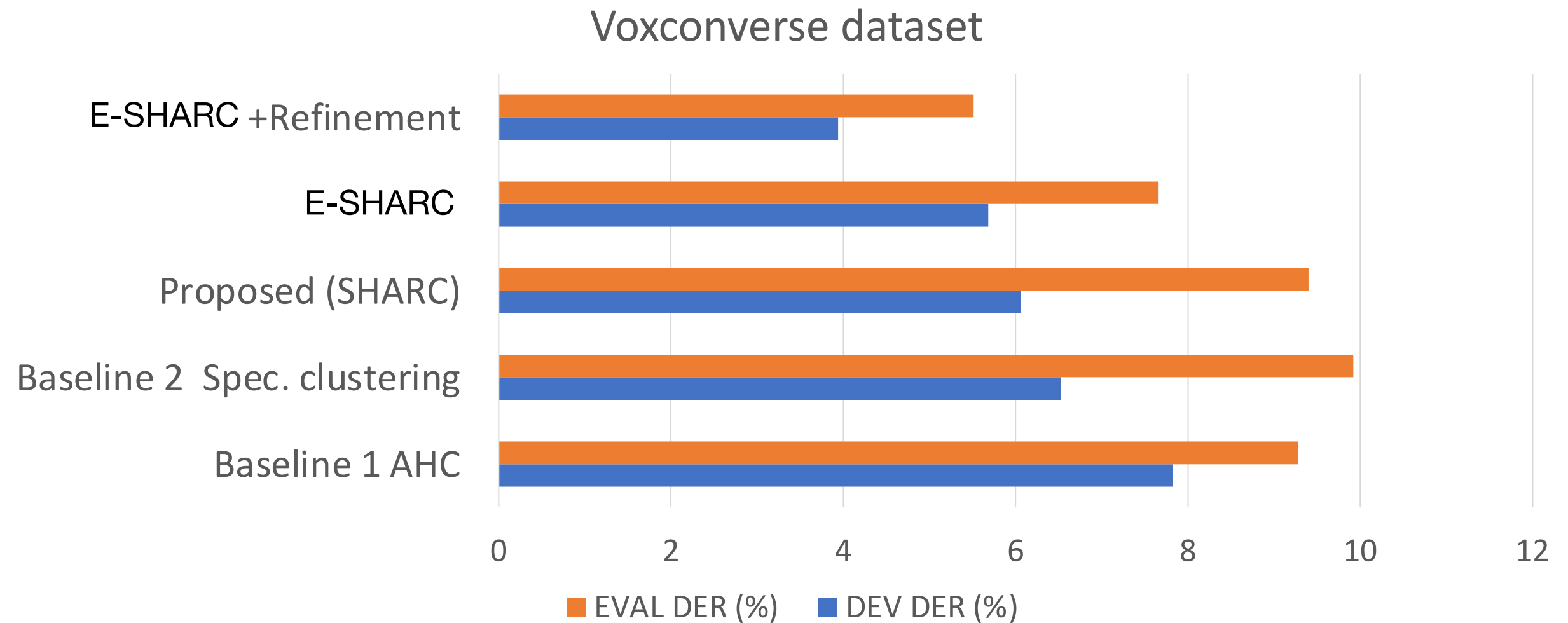


53% relative improvement over best baseline

P. Singh, A. Kaul and S. Ganapathy, "Supervised Hierarchical Clustering using Graph Neural Networks for Speaker Diarization", IEEE ICASSP 2023.

# Results

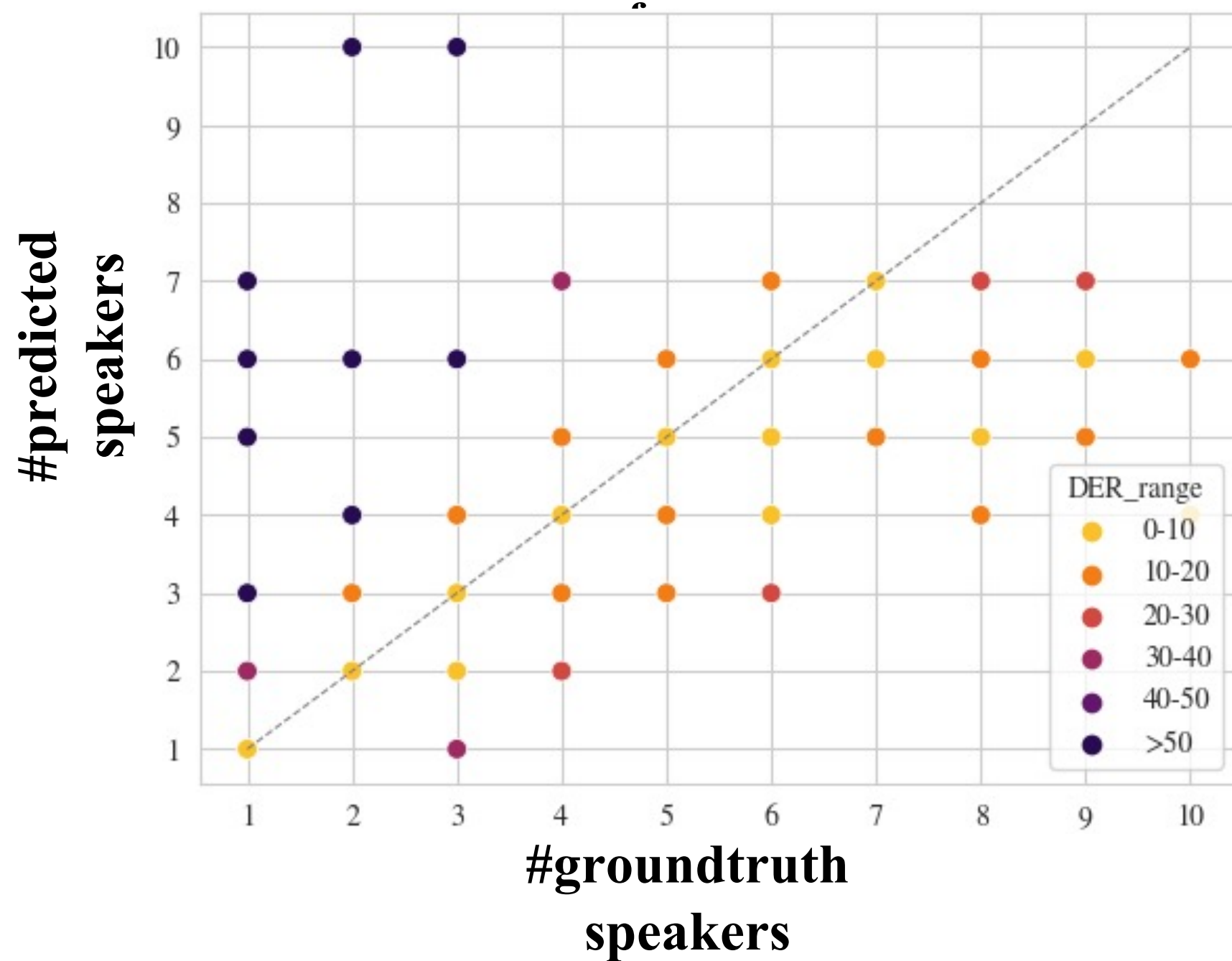
- Performance : DER (%) (lower the better)



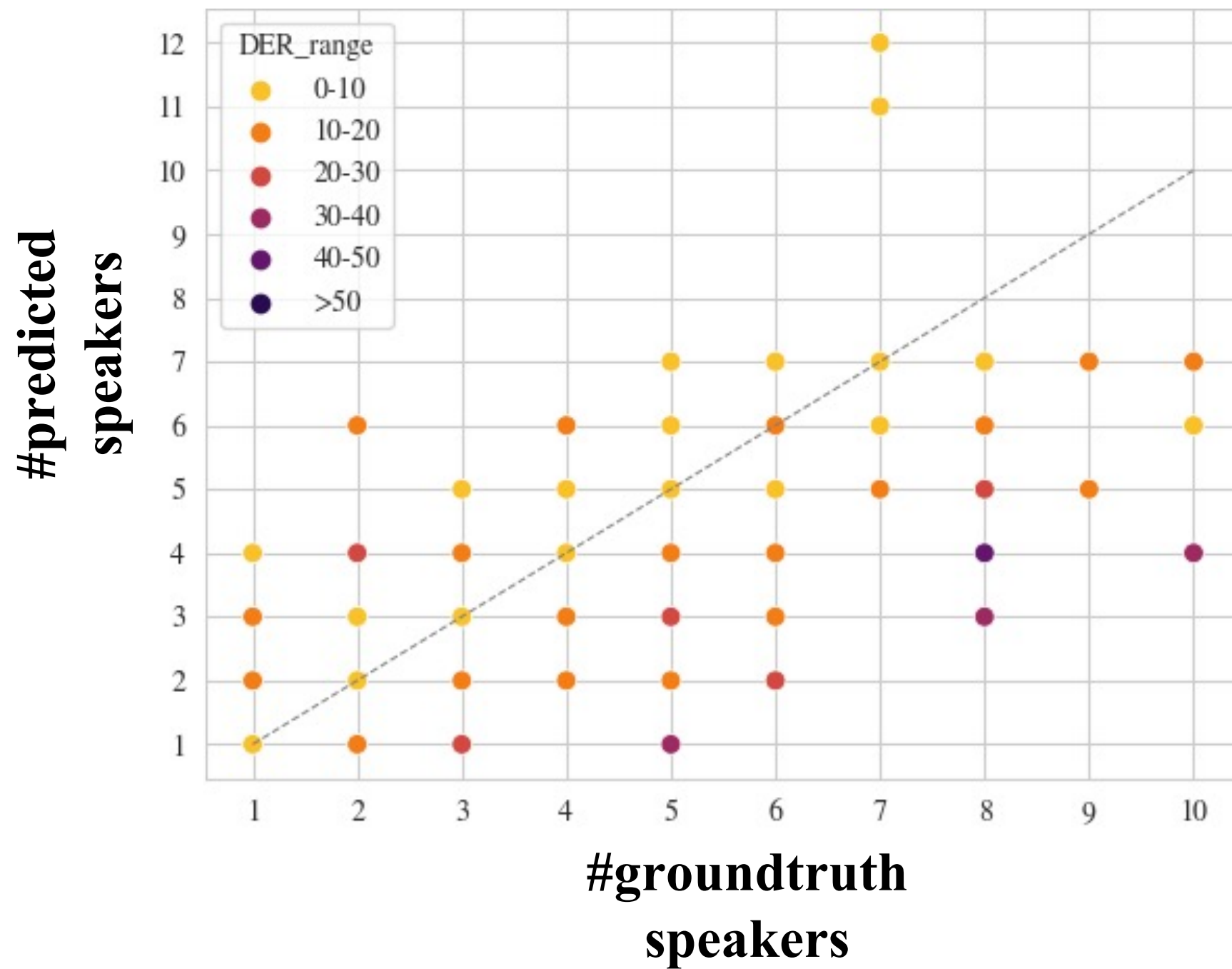
**41% relative improvement over best baseline**

## Voxconverse Dev set

(a) SC Average DER (%)



(b) ESHARC Average DER (%) performance



# Overlap Results

DER\* - with overlap + no collar

Overlap detector: Bredin et al., pyannote.audio: neural building blocks for speaker diarization, 2020

| AMI               | Eval DER* (%) |
|-------------------|---------------|
| AHC + overlap     | 26.67         |
| SC + overlap      | 20.36         |
| SHARC + overlap   | 19.50         |
| E-SHARC + overlap | 17.99         |
| Voxconverse       | Eval DER* (%) |
| AHC + overlap     | 12.05         |
| SC + overlap      | 13.73         |
| SHARC + overlap   | 12.56         |
| E-SHARC + overlap | 11.42         |
| DISPLACE          | Eval DER* (%) |
| AHC + overlap     | 40.47         |
| SC + overlap      | 40.65         |
| SHARC + overlap   | 32.73         |
| E-SHARC + overlap | 32.45         |

# Recent works Results

- DER – without overlap + 0.25s collar
- DER\* - with overlap + no collar

| AMI SDM System            | DER*        | DER  |
|---------------------------|-------------|------|
| Pyannote [1]              | 29.1        | -    |
| x-vec+AHC+VBx [2]         | 27.4        | 12.6 |
| SelfSup-PLDA-PIC +VBx [3] | 23.8        | 5.5  |
| Raj et al. [4]            | 23.7        | -    |
| Plaquet et al. [5]        | 22.9        | -    |
| GAE-based+ SC [6]         | -           | 5.5  |
| GADEC-based [6]           | -           | 4.2  |
| E-SHARC (proposed)        | 19.83       | 2.9  |
| E-SHARC-Ovp +VBx (prop.)  | <b>17.2</b> | 2.6  |

[1] Bredin et al., Interspeech, 2021

[2] Landini et al., 2020

[3] Singh et al., ASRU, 2021

[4] Raj et al., arxiv, 2022

[5] Plaquet et al., Interspeech, 2023

[6] Wei et al., Speech Communications, 2023

# Recent works Results

- DER – without overlap + 0.25s collar
- DER\* - with overlap + no collar

| <b>Voxconverse System</b>               | DER*        | DER  |
|-----------------------------------------|-------------|------|
| Pyannote [1]                            | 11.9        | -    |
| Plaquet et al. [2]                      | 10.4        | -    |
| GAE-based+ SC [3]                       | -           | 8.0  |
| GADEC-based [3]                         | -           | 7.6  |
| E-SHARC (prop.)                         | 11.68       | 7.6  |
| E-SHARC-Ovp +VBx (proposed)             | <b>10.1</b> | 6.3  |
| <b>DISPLACE System</b>                  |             |      |
| DISPLACE Baseline [4]                   | 32.2        | 14.6 |
| E-SHARC-Ovp +VBx (prop.) + Baseline SAD | 31.4        | 13.0 |
| Winning system [4]                      | <b>27.8</b> | 7.3  |

[1] Bredin, Interspeech, 2021

[2] Plaquet et al., Interspeech, 2023

[3] Wei et al., Speech Communications, 2023

[4] Baghel et al., 2023

# Summary

---

- Introduced **supervised hierarchical clustering** for speaker diarization for the first time.
- Designed an **end-to-end approach** to perform speaker diarization using **Graph Neural Networks**.
- Introduced **overlapped speaker prediction**.
- Achieved state-of-the-art performance on benchmark datasets.

# Conclusion and Future Directions

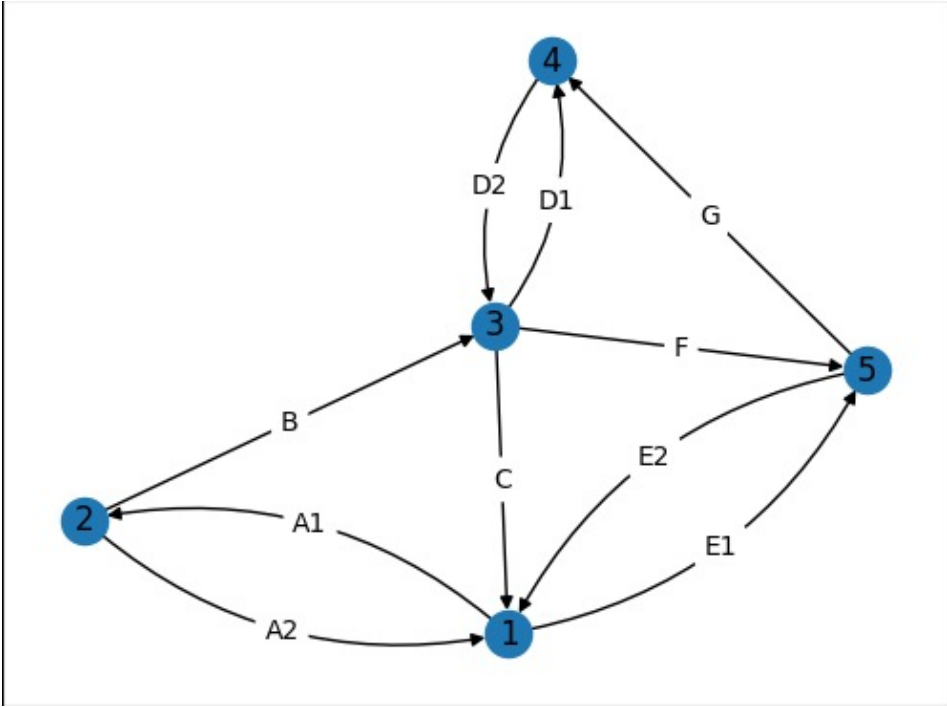
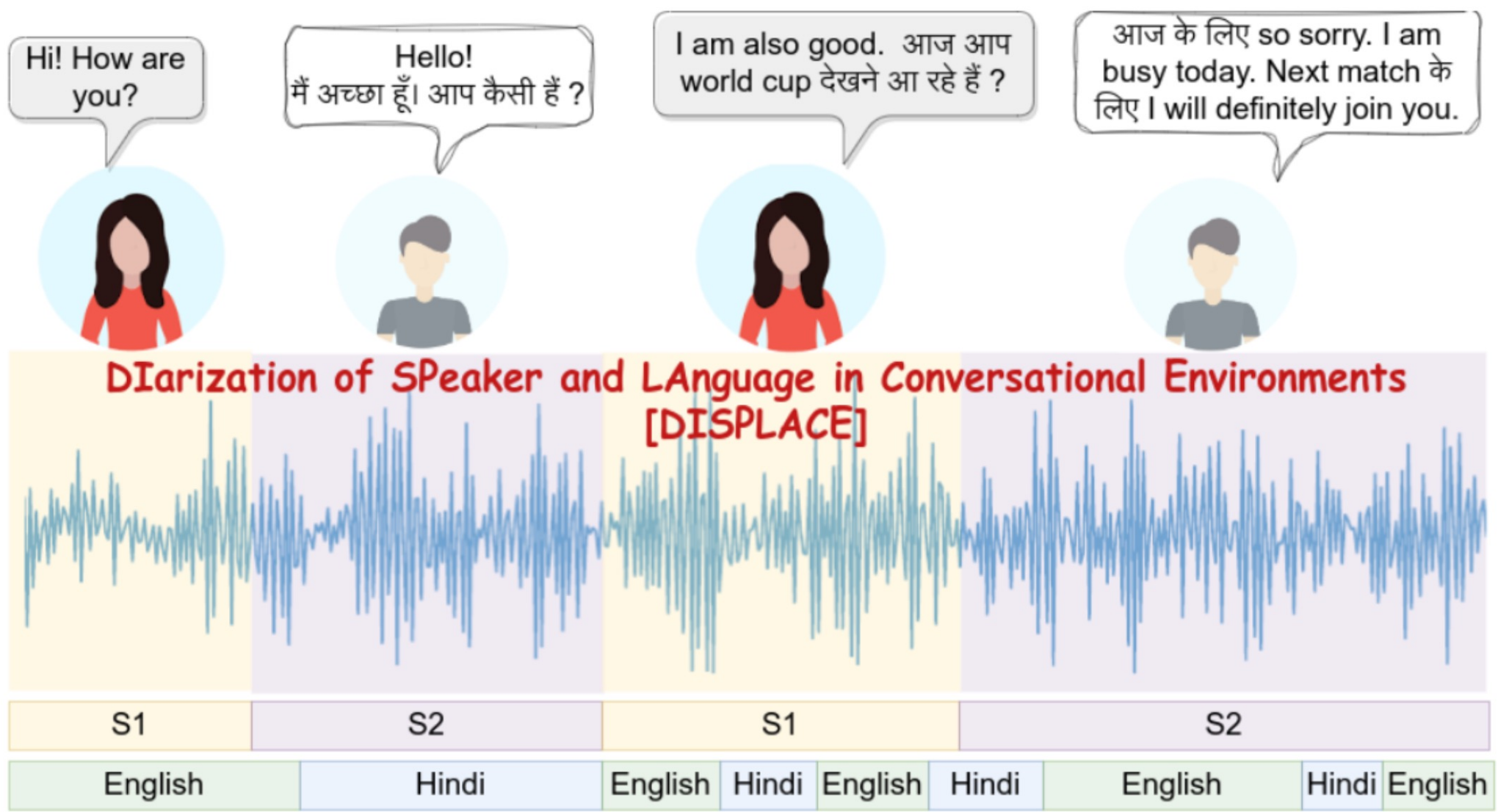
---

# Concluding remarks

| Proposed Approaches     | Novelties                                                                                                                                                                                        | Limitations                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SSC</b>              | <ul style="list-style-type: none"><li>Introduced <b>self-supervised clustering using DNN</b></li><li>Introduced <b>PIC graph clustering</b> for the first time to improve diarization.</li></ul> | <ul style="list-style-type: none"><li>Similarity scoring is not learnable (cosine)</li></ul>                                                   |
| <b>SelfSup-PLDA-PIC</b> | <ul style="list-style-type: none"><li>Introduced self-supervised <b>metric learning</b></li></ul>                                                                                                | <ul style="list-style-type: none"><li>Performance depends on initial clustering</li><li>Degrades with higher number of speakers</li></ul>      |
| <b>SHARC</b>            | <ul style="list-style-type: none"><li><b>First time</b> performed <b>supervised hierarchical clustering</b> for diarization</li></ul>                                                            | <ul style="list-style-type: none"><li>Increased training time</li><li>Require domain specific training</li><li>Not purely end-to-end</li></ul> |

# Future Directions

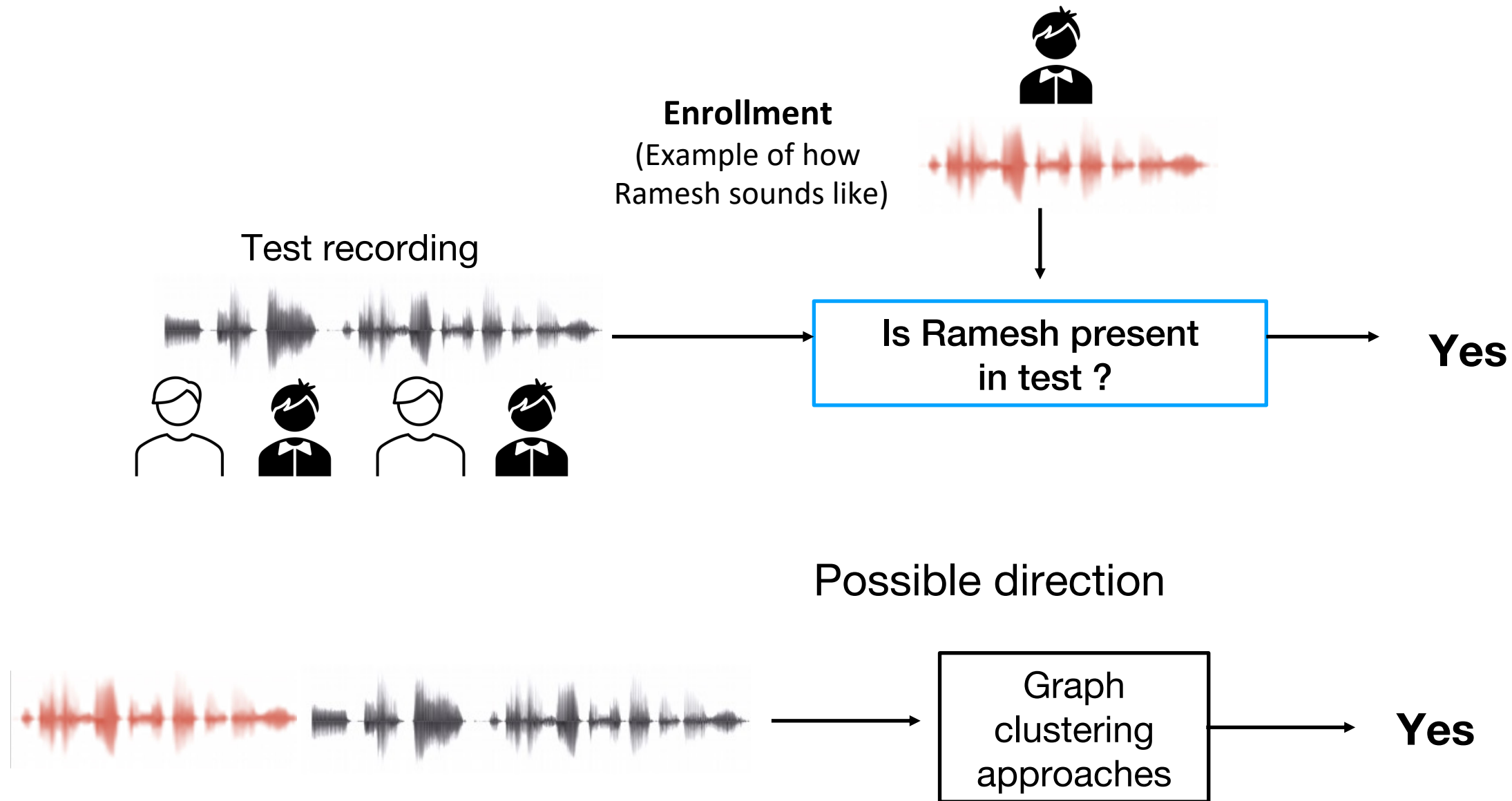
## Multilingual conversation Diarization



Use Multi-edge graph to perform multi-task learning

# Future Directions

## Target speaker identification in conversational speech



- Need to handle channel mismatch
- Avoid clustering within target speaker recording

# Publications based on the thesis

- **Peer-reviewed Journals:**

1. **P. Singh** and S. Ganapathy, “Self-supervised Representation Learning With Path Integral Clustering For Speaker Diarization”, IEEE/ACM Transactions on Audio, Speech, and Language Processing (2021).
2. **P. Singh** and S. Ganapathy, “Speaker Diarization with Graph Based Supervised Hierarchical Clustering” (under review).

- **Peer-reviewed Conferences:**

1. **P. Singh**, A. Kaul and S. Ganapathy, “Supervised Hierarchical Clustering using Graph Neural Networks for Speaker Diarization”, **IEEE ICASSP 2023**.
2. **P. Singh** and S. Ganapathy, “Self-Supervised Metric Learning with Graph Clustering for Speaker Diarization”, **IEEE ASRU 2021**.
3. **P. Singh**, R. Varma, V. Krishnamohan, S. R. Chetupalli, and S. Ganapathy. “LEAP Submission for the Third DIHARD Diarization Challenge”, **INTERSPEECH 2021**.
4. **P. Singh** and S. Ganapathy, “Deep Self-Supervised Hierarchical Clustering for Speaker Diarization”, **INTERSPEECH 2020**.
5. **P. Singh**, Harsha Vardhan MA, S. Ganapathy, A. Kanagasundaram, “LEAP Diarization System for the Second DIHARD Challenge”, **INTERSPEECH 2019**.

# Indian Institute of Science

*"Since 1909, when it came to be,  
thousands have drunk in its glory;*

*Getting in is tough but  
leaving it is more rough,  
such is the charm of IISc."*

*- Anonymous*

**Thank you for your attention !**

